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NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

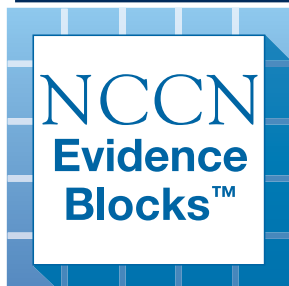
# Gestational Trophoblastic Neoplasia

## NCCN Evidence Blocks™

Version 2.2026 — November 21, 2025

NCCN.org

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available.  
Trials should be designed to maximize inclusiveness and broad representative enrollment.





# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

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# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

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## NCCN GTN Subcommittee Members

## NCCN Evidence Blocks Definitions (EB-DEF)

## Hydatidiform Mole

### Workup, Initial Treatment, Monitoring, Findings and Additional Evaluation (HM-1)

## Persistent Post-Molar GTN (HM-2)

## Principles of Imaging (HM-A)

## Gestational Trophoblastic Neoplasia (GTN)

### Workup (GTN-1)

### Primary Treatment for Low-Risk GTN (GTN-2)

### Treatment for Persistent GTN (GTN-3)

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### Primary Treatment for Intermediate Trophoblastic Tumor: PSTT and ETT (GTN-5)

## Principles of Pathology (GTN-A)

## Principles of Imaging (GTN-B)

## Principles of Surgical Management of GTN (GTN-C)

## Systemic Therapy for GTN (GTN-D)

## Principles of Gynecologic Survivorship (GTN-E)

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## Staging (ST-1)

## Abbreviations (ABBR-1)

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**NCCN Categories of Evidence and Consensus:** All recommendations are category 2A unless otherwise indicated.

See [NCCN Categories of Evidence and Consensus](#).

**NCCN Categories of Preference:**  
All recommendations are considered appropriate.

See [NCCN Categories of Preference](#).



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### NCCN EVIDENCE BLOCKS CATEGORIES AND DEFINITIONS

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| 5 |  |  |  |  |  |
| 4 |  |  |  |  |  |
| 3 |  |  |  |  |  |
| 2 |  |  |  |  |  |
| 1 |  |  |  |  |  |

E S Q C A

E = Efficacy of Regimen/Agent  
S = Safety of Regimen/Agent  
Q = Quality of Evidence  
C = Consistency of Evidence  
A = Affordability of Regimen/Agent

#### Example Evidence Block

|   |  |  |  |  |  |
|---|--|--|--|--|--|
| 5 |  |  |  |  |  |
| 4 |  |  |  |  |  |
| 3 |  |  |  |  |  |
| 2 |  |  |  |  |  |
| 1 |  |  |  |  |  |

E S Q C A

E = 4  
S = 4  
Q = 3  
C = 4  
A = 3

#### Efficacy of Regimen/Agent

|   |                                                                                                          |
|---|----------------------------------------------------------------------------------------------------------|
| 5 | <b>Highly effective:</b> Cure likely and often provides long-term survival advantage                     |
| 4 | <b>Very effective:</b> Cure unlikely but sometimes provides long-term survival advantage                 |
| 3 | <b>Moderately effective:</b> Modest impact on survival, but often provides control of disease            |
| 2 | <b>Minimally effective:</b> No, or unknown impact on survival, but sometimes provides control of disease |
| 1 | <b>Palliative:</b> Provides symptomatic benefit only                                                     |

#### Safety of Regimen/Agent

|   |                                                                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | <b>Usually no meaningful toxicity:</b> Uncommon or minimal toxicities; no interference with activities of daily living (ADLs)                   |
| 4 | <b>Occasionally toxic:</b> Rare significant toxicities or low-grade toxicities only; little interference with ADLs                              |
| 3 | <b>Mildly toxic:</b> Mild toxicity that interferes with ADLs                                                                                    |
| 2 | <b>Moderately toxic:</b> Significant toxicities often occur but life threatening/fatal toxicity is uncommon; interference with ADLs is frequent |
| 1 | <b>Highly toxic:</b> Significant toxicities or life threatening/fatal toxicity occurs often; interference with ADLs is usual and severe         |

Note: For significant chronic or long-term toxicities, score decreased by 1

#### Quality of Evidence

|   |                                                                                                  |
|---|--------------------------------------------------------------------------------------------------|
| 5 | <b>High quality:</b> Multiple well-designed randomized trials and/or meta-analyses               |
| 4 | <b>Good quality:</b> One or more well-designed randomized trials                                 |
| 3 | <b>Average quality:</b> Low quality randomized trial(s) or well-designed non-randomized trial(s) |
| 2 | <b>Low quality:</b> Case reports or extensive clinical experience                                |
| 1 | <b>Poor quality:</b> Little or no evidence                                                       |

#### Consistency of Evidence

|   |                                                                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------|
| 5 | <b>Highly consistent:</b> Multiple trials with similar outcomes                                                                    |
| 4 | <b>Mainly consistent:</b> Multiple trials with some variability in outcome                                                         |
| 3 | <b>May be consistent:</b> Few trials or only trials with few patients, whether randomized or not, with some variability in outcome |
| 2 | <b>Inconsistent:</b> Meaningful differences in direction of outcome between quality trials                                         |
| 1 | <b>Anecdotal evidence only:</b> Evidence in humans based upon anecdotal experience                                                 |

#### Affordability of Regimen/Agent (includes drug cost, supportive care, infusions, toxicity monitoring, management of toxicity)

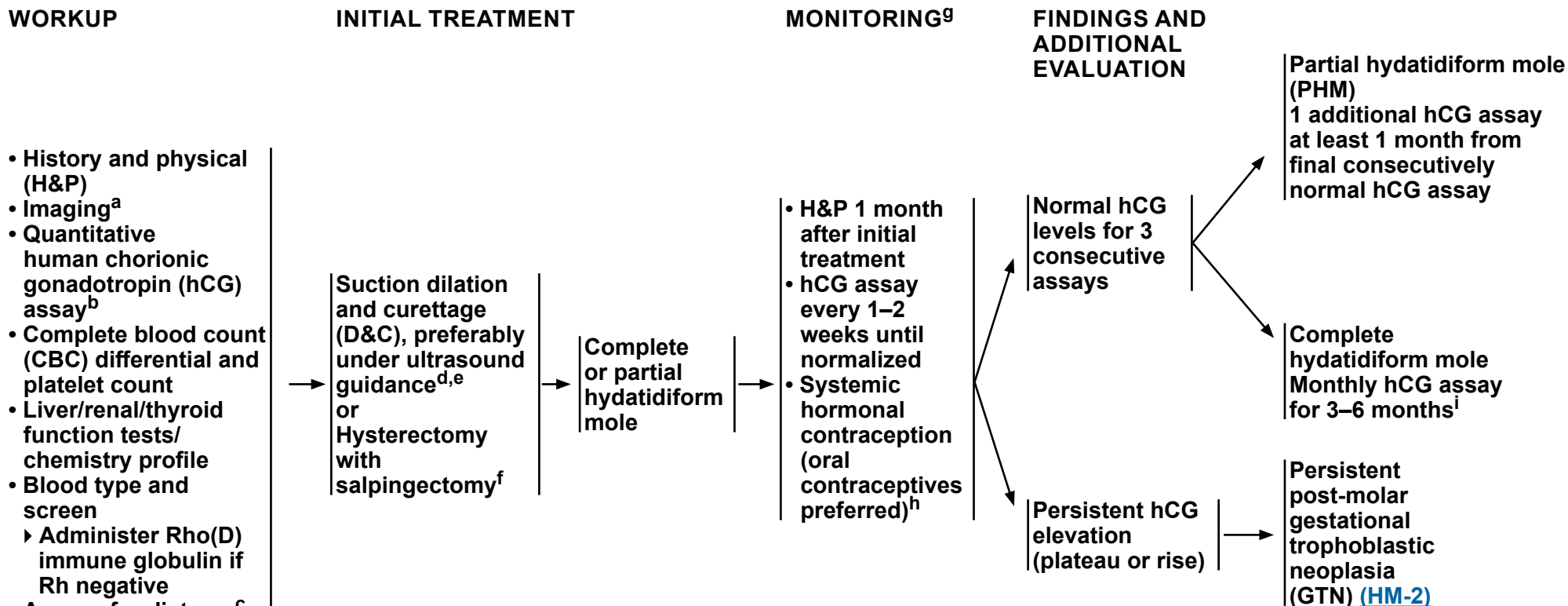
|   |                             |
|---|-----------------------------|
| 5 | <b>Very inexpensive</b>     |
| 4 | <b>Inexpensive</b>          |
| 3 | <b>Moderately expensive</b> |
| 2 | <b>Expensive</b>            |
| 1 | <b>Very expensive</b>       |



# NCCN Guidelines Version 2.2026

## Hydatidiform Mole

### NCCN Evidence Blocks™



<sup>a</sup> [Principles of Imaging \(HM-A\)](#).

<sup>b</sup> If hCG is elevated with no evidence of disease on imaging, consider possibility of luteinizing hormone (LH) crossover, pituitary hCG, or phantom hCG. Consult with laboratory medicine/pathology to test for phantom hCG with serial dilution study or comparison of serum to urine hCG.

<sup>c</sup> Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See [NCCN Guidelines for Distress Management \(DIS-A\)](#).

<sup>d</sup> Use largest curette feasible. Sharp curettage after suction. Use uterotonic drugs after initiating evacuation of uterus. Consider adding tranexamic acid (TXA) if necessary. Oxytocin receptors may be absent.

<sup>e</sup> Prophylactic chemotherapy with methotrexate or dactinomycin may be considered at the time of evacuation of a hydatidiform mole in patients at high risk for post-molar GTN (age >40 years, hCG >100,000 mIU/mL, excessive uterine enlargement, and theca lutein cysts >6 cm) when hCG follow-up is unavailable or unreliable (Wang Q, et al. Cochrane Database Syst Rev 2017;9:CD007289).

<sup>f</sup> Hysterectomy with salpingectomy may be considered as initial treatment for hydatidiform mole in patients who do not wish to preserve fertility.

<sup>g</sup> A formal follow-up program allows for early detection of GTN and limits exposure to combination chemotherapy (Sita-Lumsden A, et al. Br J Cancer 2012;107:1810-1814). See [Principles of Pathology \(GTN-A\)](#).

<sup>h</sup> Oral contraceptives are preferred over intrauterine devices (IUDs) because they suppress endogenous LH/follicle-stimulating hormone (FSH), which may interfere with hCG measurement at low levels.

<sup>i</sup> If time from D&C to hCG normalization is <56 days, consider 3 months. If ≥56 days, consider 6 months.

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
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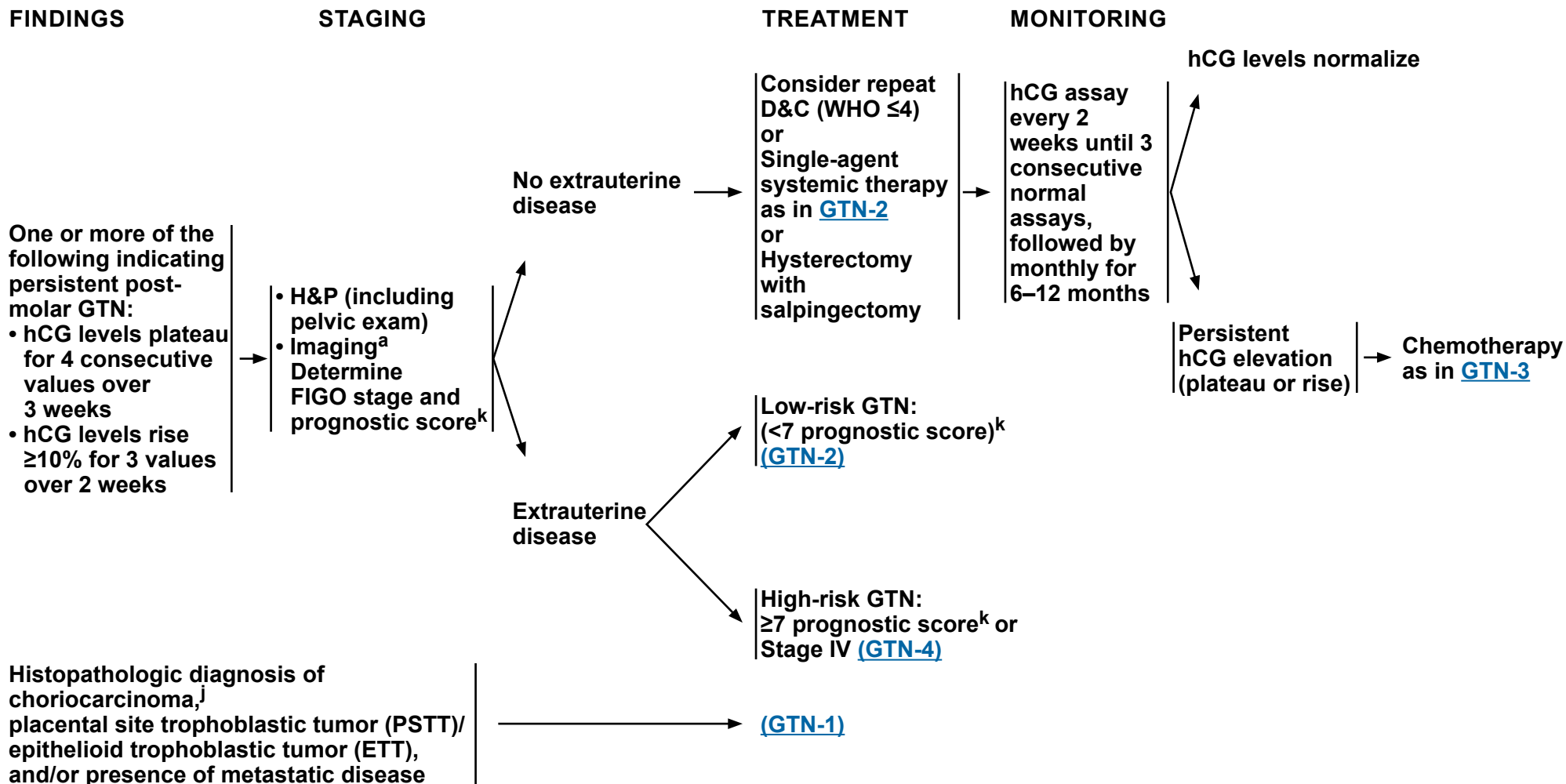




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## Hydatidiform Mole

### NCCN Evidence Blocks™



<sup>a</sup> [Principles of Imaging \(HM-A\)](#).

<sup>j</sup> [Principles of Pathology \(GTN-A\)](#).

<sup>k</sup> See [FIGO Staging System for GTN \(ST-1\)](#) and [Prognostic Scoring Index for GTN \(ST-2\)](#).

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# NCCN Guidelines Version 2.2026

## Hydatidiform Mole

### NCCN Evidence Blocks™

#### PRINCIPLES OF IMAGING

##### Initial Workup

- Hydatidiform mole

- ▶ Pelvis ultrasound
- ▶ Chest x-ray

- ◊ If the chest x-ray is positive for metastases, then perform chest/abdomen/pelvis (C/A/P) CT with contrast and brain MRI with and without contrast and manage as GTN after initial uterine evacuation.

##### Staging

- Persistent post-molar GTN

- ▶ Pelvis ultrasound

- ◊ Used to confirm absence of pregnancy, measure uterine size, and determine volume and vasculature of tumor within the uterus.

- ▶ Chest x-ray or C/A/P CT scan with contrast

- ◊ If the chest x-ray is normal, no further imaging is indicated before commencing treatment. If the chest x-ray shows metastases, C/A/P CT scan with contrast and brain MRI with and without contrast are indicated.

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**WORKUP**  
(unless previously done)

## FINDINGS<sup>h</sup>

## TREATMENT

- H&P<sup>a</sup>
- Repeat CBC differential and platelet count
- Repeat liver/renal/thyroid function tests/chemistry profile
- Imaging<sup>b</sup>
- hCG assay<sup>c,d</sup>
- Determine FIGO stage and prognostic score<sup>e</sup>
- Histopathologic assessment<sup>f</sup>
- Assess for distress<sup>g</sup>

**Low-risk GTN**  
(**<7 prognostic score**)<sup>e</sup>

➤ (GTN-2)

**High-risk GTN:  
≥7 prognostic score<sup>e</sup>  
or  
Stage IV**

➤ (GTN-4)

- Intermediate trophoblastic tumor<sup>i</sup>
  - PSTT
  - ETT

➤ **(GTN-5)**

<sup>a</sup> If visible lesions are seen in lower genital tract, do NOT biopsy due to risk of hemorrhage.

b Principles of Imaging (GTN-B).

c If hCG is elevated with no evidence of disease on imaging, consider the possibility of LH crossover or phantom hCG. Consult with laboratory medicine/pathology to test for phantom hCG with serial dilution study or comparison of serum to urine hCG.

<sup>d</sup> If hCG is elevated, although at a low level that is stable over time (and false positive hCG ruled out), quiescent GTN is possible, and close follow-up, rather than immediate treatment may be considered.

<sup>e</sup> See [FIGO Staging System for GTN \(ST-1\)](#) and [Prognostic Scoring Index for GTN \(ST-2\)](#).

<sup>f</sup> Principles of Pathology (GTN-A).

<sup>9</sup> Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See [NCCN Guidelines for Distress Management \(DIS-A\)](#).

<sup>h</sup> Consider consultation with a clinician or center with expertise in management of gestational trophoblastic diseases.

i Prognostic scoring is not valid for intermediate tumors.

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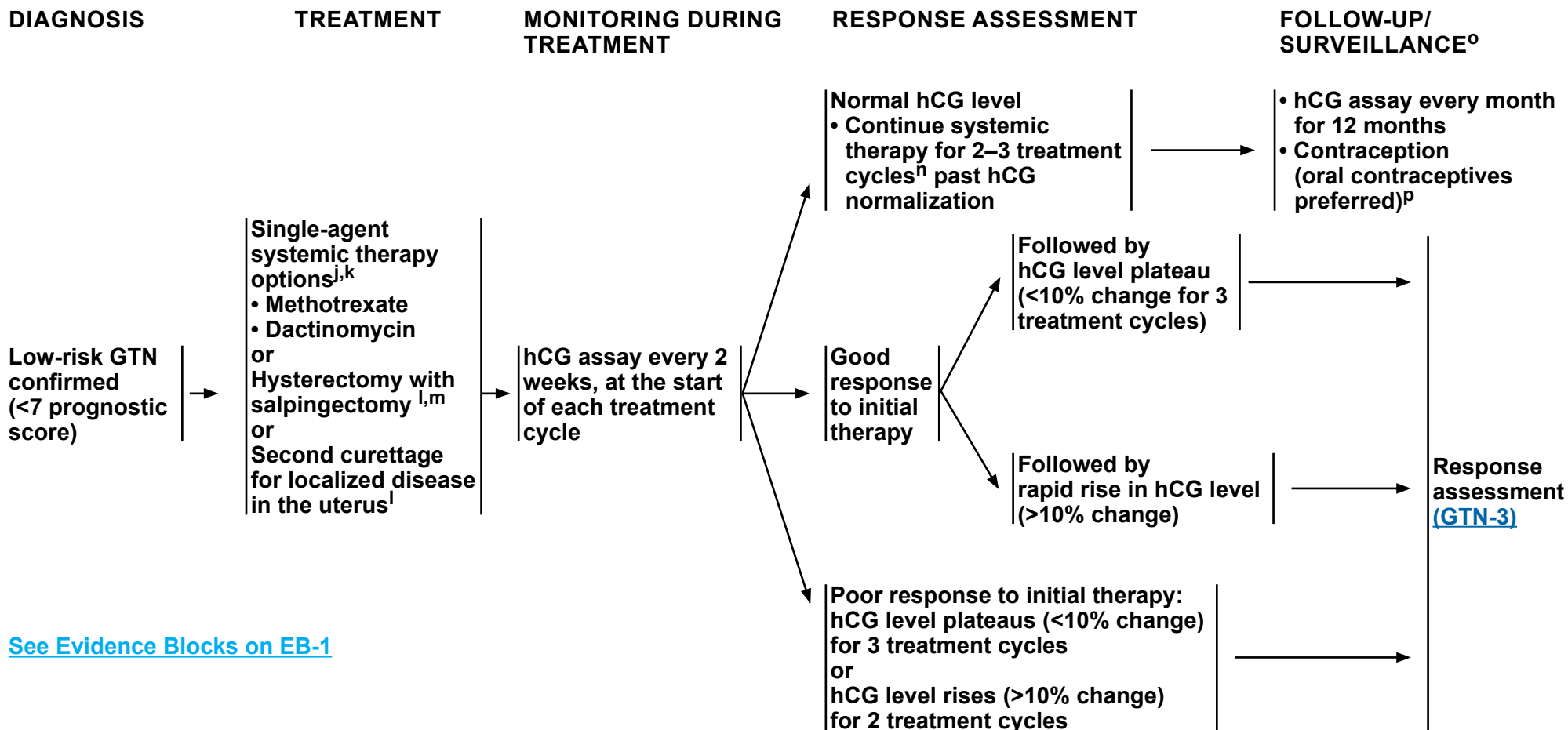


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## Gestational Trophoblastic Neoplasia

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#### PRIMARY TREATMENT FOR LOW-RISK GTN



[See Evidence Blocks on EB-1](#)

<sup>k</sup> See [Systemic Therapy for GTN \(GTN-D\)](#) for specific recommendations.

<sup>j</sup> Regimens are continued until 2–3 full cycles past normalization of the hCG.

<sup>l</sup> Hysterectomy with salpingectomy is preferred when fertility preservation is not desired. Leave ovaries in situ, even in presence of theca lutein cysts. See [Principles of Surgical Management \(GTN-C\)](#).

<sup>m</sup> Consider hysterectomy alone without further systemic therapy for patients without choriocarcinoma and a WHO score of <5.

<sup>n</sup> Lybol C, et al. Gynecol Oncol 2012;125:576-579.

<sup>o</sup> [Principles of Gynecologic Survivorship \(GTN-E\)](#).

<sup>p</sup> Oral contraceptives are preferred over IUDs because they suppress endogenous LH/FSH, which may interfere with hCG measurement at low levels.

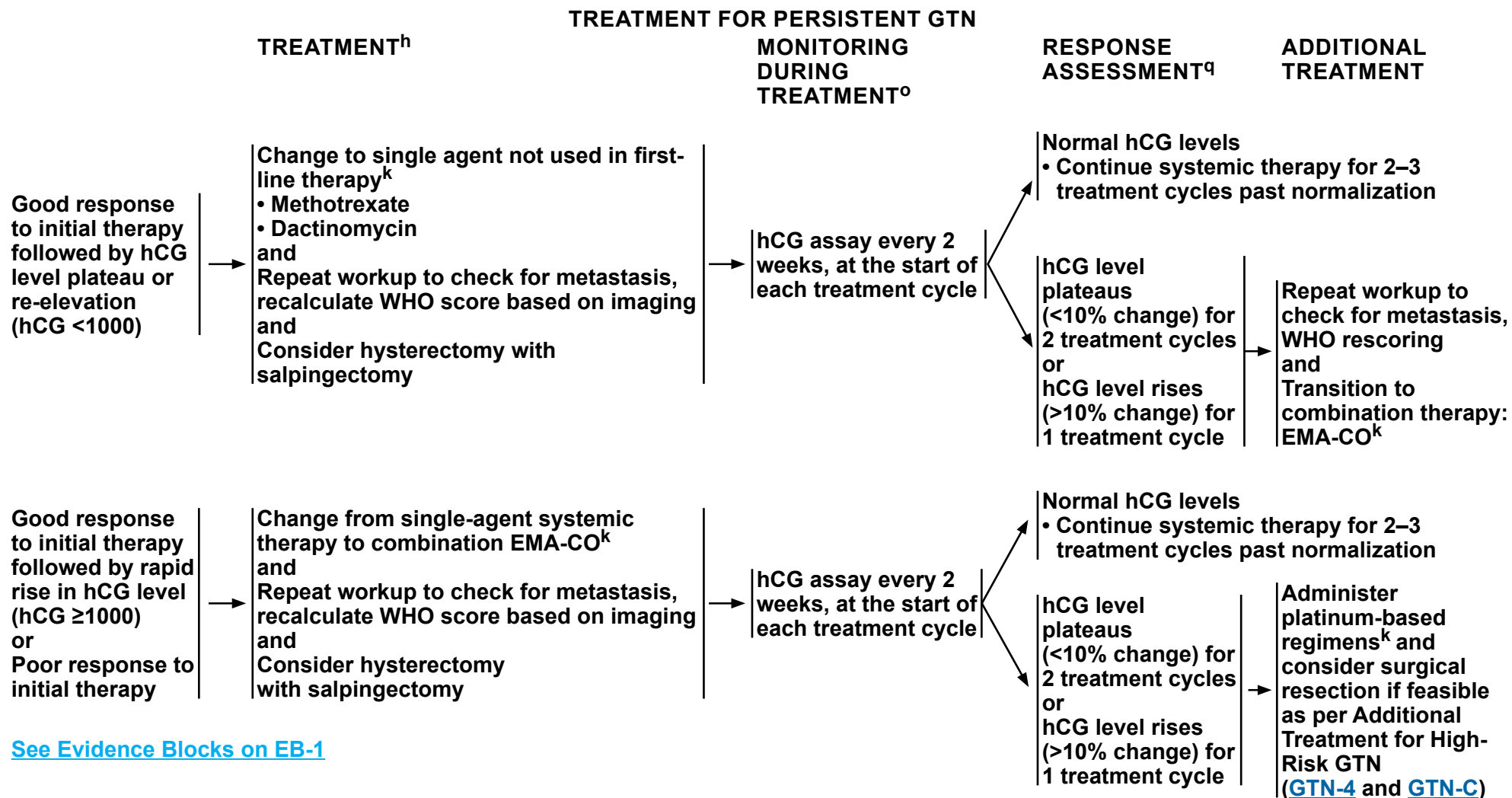
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## Gestational Trophoblastic Neoplasia

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[See Evidence Blocks on EB-1](#)

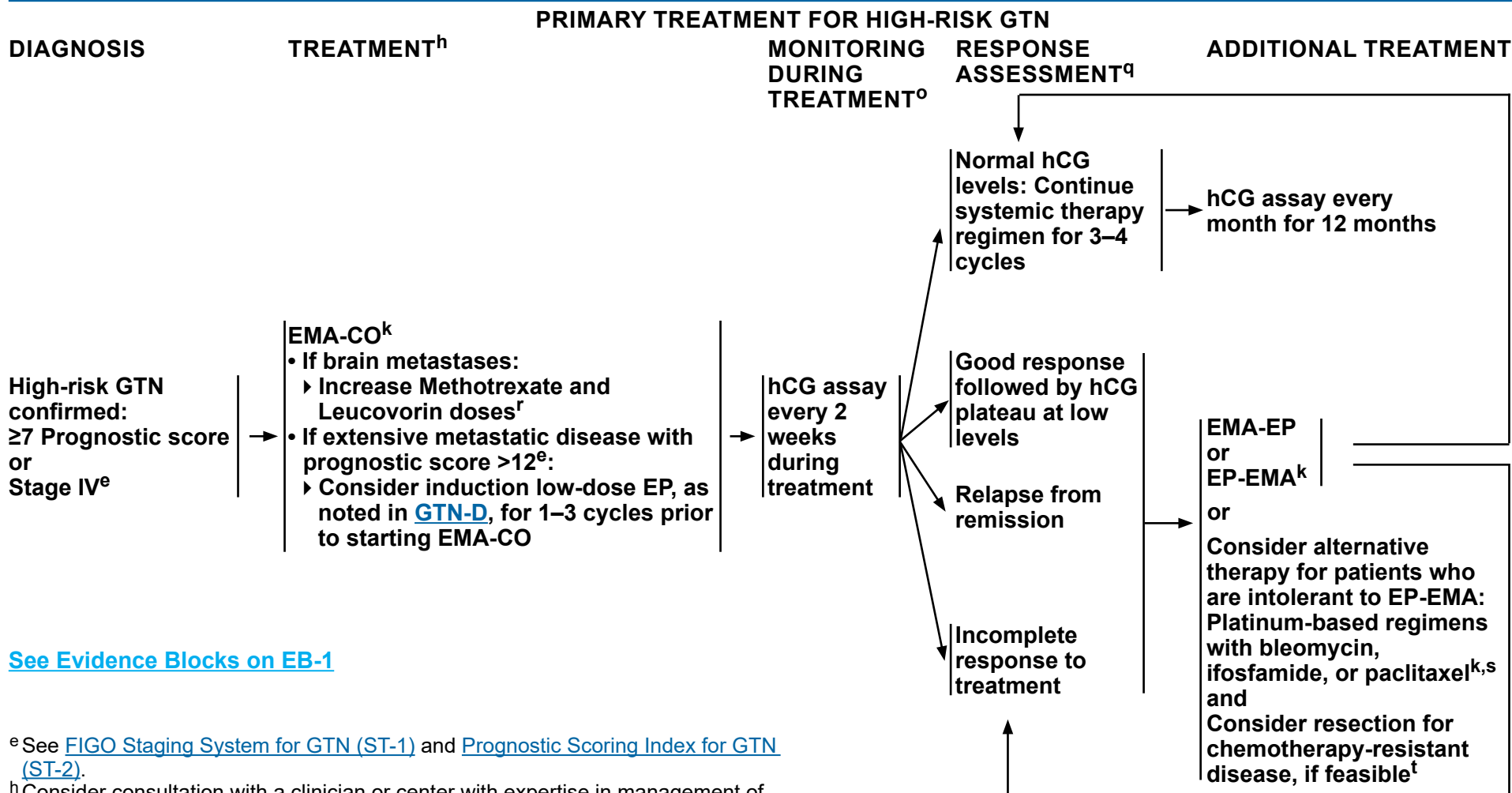
<sup>h</sup> Consider consultation with a clinician or center with expertise in management of gestational trophoblastic diseases.

<sup>k</sup> See [Systemic Therapy for GTN \(GTN-D\)](#) for specific recommendations.

<sup>o</sup> [Principles of Gynecologic Survivorship \(GTN-E\)](#).

<sup>q</sup> Post-treatment imaging is not routinely recommended for follow-up after hCG normalization in patients with post-molar GTN or choriocarcinoma, where hCG is a reliable tumor marker. It may be necessary to follow patients with brain metastases with MRI brain with contrast for 2–3 years. See [Principles of Imaging \(GTN-B\)](#).

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**See Evidence Blocks on EB-1**

<sup>e</sup> See [FIGO Staging System for GTN \(ST-1\)](#) and [Prognostic Scoring Index for GTN \(ST-2\)](#).

<sup>h</sup> Consider consultation with a clinician or center with expertise in management of gestational trophoblastic diseases.

<sup>k</sup> See [Systemic Therapy for GTN \(GTN-D\)](#) for specific recommendations.

- Principles of Gynecologic Survivorship (GTN-E).

q Post-treatment imaging is not routinely recommended for follow-up after hCG normalization in patients with post-molar GTN or choriocarcinoma, where hCG is a reliable tumor marker. It may be necessary to follow patients with brain metastases with MRI brain with contrast for 2–3 years. See [Principles of Imaging \(GTN-B\)](#).

<sup>r</sup> For dosing modifications for brain metastases, see [Systemic Therapy for GTN \(GTN-D 2 of 7\)](#)

S Additional Agents/Regimens Shown to Have Some Activity in Treating Multiagent Chemotherapy-Resistant GTN (GTN-D 6 of 7).

<sup>t</sup> Consider surgery, especially hysterectomy with salpingectomy and pulmonary resection, for chemotherapy-resistant disease. See [Principles of Surgical Management \(GTN-C\)](#).

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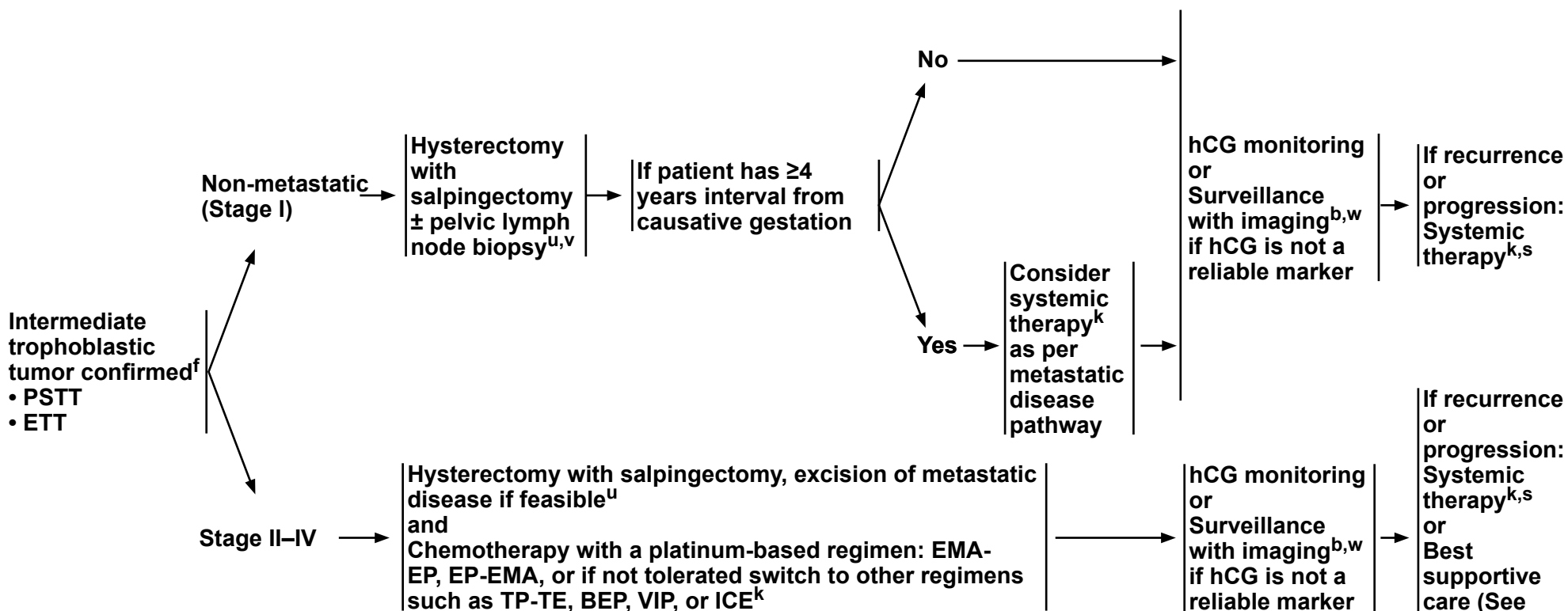
## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### DIAGNOSIS

#### PRIMARY TREATMENT FOR PSTT AND ETT TREATMENT<sup>h</sup>

#### MONITORING AND SURVEILLANCE<sup>b,o</sup>



[See Evidence Blocks on EB-1](#)

<sup>b</sup> [Principles of Imaging \(GTN-B\)](#).

<sup>f</sup> [Principles of Pathology \(GTN-A\)](#).

<sup>h</sup> Consider consultation with a clinician or center with expertise in management of gestational trophoblastic diseases.

<sup>k</sup> See [Systemic Therapy for GTN \(GTN-D\)](#) for specific recommendations.

<sup>o</sup> [Principles of Gynecologic Survivorship \(GTN-E\)](#).

<sup>s</sup> [Additional Agents/Regimens Shown to Have Some Activity in Treating Multiagent Chemotherapy-Resistant GTN \(GTN-D 6 of 7\)](#).

<sup>u</sup> [Principles of Surgical Management \(GTN-C\)](#).

<sup>v</sup> The incidence of pelvic lymph node metastasis in PSTT/ETT is estimated to be between 5% and 15% in clinical stage I tumors. Therefore, pelvic lymph node biopsy should be considered at the time of hysterectomy with salpingectomy, especially with large, deeply invasive tumors.

<sup>w</sup> Post-treatment imaging is indicated for follow-up after treatment of PSTT and ETT, where hCG is a less reliable tumor marker.

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# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### PRINCIPLES OF PATHOLOGY

##### Procedure

- D&C
- Hysterectomy
- Other

##### Pathologic Diagnosis of Benign, Noninvasive Hydatidiform Mole<sup>1</sup>

|                      | Gross Examination                                  | Chorionic Villi                                                               | Trophoblastic Hyperplasia         | Cytologic Atypia | Villous Stroma                                                                                                       | p57 IHC                                                                | DNA Genotyping <sup>a</sup>        |
|----------------------|----------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------|
| <b>Complete mole</b> | Diffuse hydropic villi; fetus absent               | Diffusely enlarged                                                            | Marked, typically circumferential | May be marked    | Marked edema with cisterns and trophoblast inclusions; blood vessels absent; nucleated red blood cells (RBCs) absent | Absent nuclear staining in cytotrophoblasts and villous stromal cells  | Diploid diandric genome            |
| <b>Partial mole</b>  | Patchy hydropic villi; fetal tissue may be present | Dual population of villi composed of enlarged villi and small, fibrotic villi | Mild                              | Mild             | May have cisterns; trophoblastic pseudoinclusions; blood vessels present; nucleated RBCs present                     | Nuclear staining present in cytotrophoblasts and villous stromal cells | Triploid diandric-monogynic genome |

- Report histologic type only.
- If myometrial or vascular invasion is present (ie, invasive mole), use pathologic assessment for malignant GTN below.

<sup>a</sup> Definitive diagnosis of a PHM is not possible without tissue short tandem repeat (STR) genotyping. Neither histologic nor immunophenotypic examination can reliably distinguish a PHM from a non-molar hydropic abortus. Karyotyping is not a recommended ancillary test, as it cannot distinguish PHMs from triploid digynic-monoandric non-molar gestations.

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# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### PRINCIPLES OF PATHOLOGY

##### **Pathologic Assessment for Malignant Gestational Trophoblastic Tumor<sup>b</sup>**

- Tumor site (ie, uterine corpus, cervix, other, cannot be determined)
- Tumor size (measured in cm)
- Histologic type
  - Hydatidiform mole, invasive
  - Choriocarcinoma
  - PSTT
  - ETT
  - Malignant trophoblastic tumor, type cannot be determined
- Tissue/organ involvement (list all organs submitted for evaluation involved by tumor)
- Specimen margin status (where applicable; when negative, may also include closest margin and distance to closest margin)
- Lymphovascular space invasion (LVSI)

##### **Immunohistochemical Markers for Differential Diagnosis of GTN<sup>1,2,3</sup>**

- GATA-3 is a sensitive marker for both benign and malignant trophoblast proliferations, and may be useful in distinguishing GTN from non-GTN tumors.

|                             | Mel-CAM (CD146) | hPL | Beta-hCG | p63 | Cyclin E | Ki-67              |
|-----------------------------|-----------------|-----|----------|-----|----------|--------------------|
| PSTT                        | +               | +   | —        | —   |          | >10%               |
| ETT                         | —               | —   | —        | +   | >50%     | >10%               |
| Gestational choriocarcinoma | +/-             | +   | +        | —   |          | High, usually >90% |

Mel-CAM: melanoma cell adhesion molecule; hPL: human placental lactogen; Beta-hCG: beta-human chorionic gonadotropin

##### **Footnotes**

<sup>b</sup>Benign trophoblastic tumors/lesions (ie, exaggerated placental site, placental site nodule) are benign and should NOT be reported using these principles.

##### **References**

<sup>1</sup> WHO Classification of Tumours Editorial Board. Female Genital Tumours WHO Classification of Tumours, 5th ed, vol 4; 2020.

<sup>2</sup> College of American Pathologists: Protocol for the Examination of Specimens From Patients With Primary Gestational Trophoblastic Malignancy. 2017. Available at: <https://documents.cap.org/protocols/cp-trophoblast-17protocol-4000.pdf>.

<sup>3</sup> Chi DS, Berchuck A, Dizon DS, et al. Table 26.2: Immunohistochemical Markers for Differential Diagnosis of GTD. Principles and Practice of Gynecologic Oncology. Lippincott Williams & Wilkins; 2017:745.

**Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#). All recommendations are category 2A unless otherwise indicated.**





# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### PRINCIPLES OF IMAGING<sup>a</sup>

##### Initial Workup

- **GTN**
  - ▶ **C/A/P CT scan with contrast**
    - ◊ If contrast is contraindicated, other imaging techniques such as MRI may be considered.
      - Brain MRI (preferred) with and without contrast or CT with contrast (if pulmonary metastasis)
  - ▶ **Pelvis ultrasound or MRI with and without contrast**

##### Staging

- **Persistent GTN**
  - ▶ **Repeat imaging workup (above) to check for metastasis and recalculate WHO score based on imaging.**

##### Follow-up/Surveillance

- **Low-risk GTN**
  - ▶ **Post-treatment imaging is not routinely recommended for follow-up after hCG normalization in patients with post-molar GTN or choriocarcinoma, where hCG is a reliable tumor marker.**
    - ◊ It may be necessary to follow patients with brain metastases with MRI brain with and without contrast for 2–3 years.
- **High-risk GTN**
  - ▶ **Post-treatment imaging is not routinely recommended for follow-up after hCG normalization in patients with post-molar GTN or choriocarcinoma, where hCG is a reliable tumor marker.**
    - ◊ It may be necessary to follow patients with brain metastases with MRI brain with and without contrast for 2–3 years.
- **Intermediate trophoblastic tumor: PSTT and ETT (non-metastatic and metastatic disease)**
  - ▶ **Post-treatment imaging is indicated for follow-up after treatment of PSTT and ETT, where hCG is a less reliable tumor marker.**
    - ◊ **Consider FDG-PET/CT (whole body) at the completion of chemotherapy and then FDG-PET/CT or CT C/A/P every 6–12 months for 2–3 years.**

<sup>a</sup> MRI is performed with and without IV contrast and CT is performed with contrast throughout the guidelines unless contraindicated. Contrast is not required for screening chest CT.

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
All recommendations are category 2A unless otherwise indicated.



- Chemotherapy is the mainstay of treatment for GTN; however, surgery plays an important role in specific clinical circumstances.
- Hysterectomy is not required in the treatment of GTN; most patients can be cured without hysterectomy.
- For patients who do not desire future fertility with low-risk disease, hysterectomy can decrease the number of cycles of chemotherapy needed to achieve remission.
- Hysterectomy can also be considered in rare cases of uterine-confined, chemo-resistant disease or in cases of hemorrhage requiring repeated transfusion.
- Hysterectomy can be performed via minimally invasive or open approach without compromise in oncologic outcomes. Ovarian removal is not necessary.
- Lung metastasis can be a site of chemotherapy-resistant disease. For these highly selected patients, resection of pulmonary metastasis can improve cure rates.
- Craniotomy can be considered in rare cases of GTN with brain metastases, including active intracranial hemorrhage, raised intracranial pressure, and impending cerebral herniation.

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
All recommendations are category 2A unless otherwise indicated.



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### SYSTEMIC THERAPY FOR GTN

##### Regimens for Low-Risk GTN

| Preferred                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Comments/Considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Methotrexate</u></b><br>0.4 mg/kg/day (max 25 mg/day) IV Push (preferred) or IM daily x 5 Days;<br>Repeat every 14 Days<br>(category 1)<br><br>OR<br><br>1 mg/kg IM every other day x 4 days (Days 1, 3, 5, and 7)<br>Alternating every other day with Leucovorin 0.1 mg/kg rounded up to the<br>nearest 5-mg dose increment or 15 mg PO (preferred) or<br>IM 30 hours after each Methotrexate dose on Days 2, 4, 6, and 8;<br>Repeat every 14 Days<br>(category 1) | <ul style="list-style-type: none"><li>• A multiday Methotrexate regimen is typically used as first-line therapy for low-risk GTN. Due to its toxicity profile, 5-Day Dactinomycin has most often been used as secondary therapy for patients with Methotrexate toxicity or effusions contradicting the use of Methotrexate.</li></ul> <b><u>NOT RECOMMENDED</u></b> <ul style="list-style-type: none"><li>• Methotrexate 30–50 mg/m<sup>2</sup> IM weekly<br/>OR<br/>Methotrexate infusion<br/>(eg, 300 mg/m<sup>2</sup> over 12 hours/Leucovorin) due to lesser efficacy.</li></ul> |
| <b><u>Dactinomycin</u></b><br>Preferred dosing option: 0.5 mg fixed dose IV Push daily x 5 Days;<br>Repeat every 14 Days<br>(category 1)<br><br>OR<br><br>1.25 mg/m <sup>2</sup> (max 2 mg) IV Push pulse;<br>Repeat every 14 Days<br>(category 1)                                                                                                                                                                                                                        | <ul style="list-style-type: none"><li>• Dactinomycin pulse regimen should not be used as secondary therapy for Methotrexate-resistant disease nor as primary therapy in patients with choriocarcinoma.</li></ul>                                                                                                                                                                                                                                                                                                                                                                     |

[Continued](#)

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
All recommendations are category 2A unless otherwise indicated.



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### SYSTEMIC THERAPY FOR GTN

##### High-Risk GTN: Primary Therapy Options<sup>a,b</sup>

| Preferred                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Comments/Considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EMA-CO (Etoposide/Methotrexate/Dactinomycin-Cyclophosphamide/Vincristine)</b><br><b>(Repeat every 2 Weeks until hCG normalizes, then continue for an additional 6–8 weeks)</b> <ul style="list-style-type: none"><li>• Etoposide 100 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1 and 2</li><li>• Dactinomycin 0.5 mg IV Push on Days 1 and 2</li><li>• Methotrexate 300 mg/m<sup>2</sup> IV infusion over 12 hours on Day 1<ul style="list-style-type: none"><li>▶ Leucovorin 15 mg PO (preferred) or IM every 12 hours for 4 doses starting 24 Hours after the start of Methotrexate infusion</li></ul></li><li>• Cyclophosphamide 600 mg/m<sup>2</sup> IV over 30 minutes on Day 8</li><li>• Vincristine 0.8 mg/m<sup>2</sup> (maximum of 2 mg) IV over 5–10 minutes on Day 8</li></ul> | <ul style="list-style-type: none"><li>• Consider low-dose induction chemotherapy with<ul style="list-style-type: none"><li>▶ Etoposide 100 mg/m<sup>2</sup>/day IV and Cisplatin 20 mg/m<sup>2</sup>/day IV on Days 1 and 2 every 7 Days for 1–3 courses prior to starting EMA-CO in patients with widely metastatic disease (prognostic score &gt;12) who are at significant risk for pulmonary, intraperitoneal, or intracranial hemorrhage.</li></ul></li><li>• For secondary prophylaxis of neutropenic fever, or for treatment delay:<ul style="list-style-type: none"><li>▶ Filgrastim, 5 mcg/kg 3–4 days/wk [ie, on Days 4–6(7) and 10–12(13)] of each EMA-CO cycle.<sup>c</sup></li></ul></li><li>• For patients with brain metastases, increase the Methotrexate infusion dose in the EMA-CO protocol to 1000 mg/m<sup>2</sup> and extend the infusion from 12 to 24 hours; give Leucovorin 15 mg PO every 6 hours for 12 doses starting 32 hours after the start of the Methotrexate infusion.</li></ul> |

<sup>a</sup> FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.

<sup>b</sup> Use granulocyte colony-stimulating factor (G-CSF) as primary prophylaxis with etoposide/cisplatin or etoposide/carboplatin-based regimens. See [NCCN Guidelines for Hematopoietic Growth Factors](#).

<sup>c</sup> Round filgrastim dose to the nearest vial size by institution-defined weight limits.

**Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#). All recommendations are category 2A unless otherwise indicated.**

**Continued**

**GTN-D  
2 OF 7**

## SYSTEMIC THERAPY FOR GTN

**High-Risk GTN: Primary Therapy Options<sup>a,b</sup> – Continued**

**For patients at highest risk, consider: EMA-EP or EP-EMA**

- **EMA-EP (or EP-EMA) regimen is considered the most appropriate therapy for patients whose disease has responded to EMA-CO but have plateauing low hCG levels or have developed re-elevation of hCG levels after a complete response to EMA-CO.**

| Preferred                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comments/Considerations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b><u>EMA-EP (Etoposide/Methotrexate/Dactinomycin- Etoposide/Cisplatin)<sup>d</sup></u></b><br/> <b>(Repeat every 2 Weeks alternating EMA with EP weekly through 6–8 Weeks post-serologic remission)</b></p> <ul style="list-style-type: none"> <li>• Etoposide 100 mg/m<sup>2</sup>/day IV over 60 minutes on Day 1</li> <li>• Methotrexate 300 mg/m<sup>2</sup> IV infusion over 12 hours on Day 1               <ul style="list-style-type: none"> <li>▶ Leucovorin 15 mg PO (preferred) or IM every 12 hours for 4 doses beginning 24 hours after the start of the Methotrexate infusion</li> </ul> </li> <li>• Dactinomycin 0.5 mg IV push on Day 1</li> </ul> | <ul style="list-style-type: none"> <li>• Consider low-dose induction chemotherapy with               <ul style="list-style-type: none"> <li>▶ Etoposide 100 mg/m<sup>2</sup>/day IV and Cisplatin 20 mg/m<sup>2</sup>/day IV on Days 1 and 2 every 7 Days for 1–3 courses prior to starting EMA-EP or EP-EMA in patients with widely metastatic disease (prognostic score &gt;12) who are at significant risk for pulmonary, intraperitoneal, or intracranial hemorrhage.</li> </ul> </li> <li>• For patients with brain metastases, increase the Methotrexate infusion dose in the EP-EMA or EMA-EP protocol to 1000 mg/m<sup>2</sup> and extend the infusion from 12 to 24 hours; give Leucovorin 15 mg PO every 6 hours for 12 doses starting 32 hours after the start of the Methotrexate infusion.</li> </ul> |
| <p><b><u>EP-EMA (Etoposide/Cisplatin-Etoposide/Methotrexate/Dactinomycin)<sup>d</sup></u></b><br/> <b>(Repeat every 2 Weeks alternating EP with EMA weekly through 6–8 Weeks post-serologic remission)</b></p> <ul style="list-style-type: none"> <li>• Etoposide 100–150 mg/m<sup>2</sup> IV over 60 minutes on Day 1</li> <li>• Cisplatin 75 mg/m<sup>2</sup> IV over 2 hours on Day 1</li> </ul>                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <ul style="list-style-type: none"> <li>• Etoposide 100 mg/m<sup>2</sup> IV over 60 minutes on Day 8</li> <li>• Methotrexate 300 mg/m<sup>2</sup> IV infusion over 12 hours on Day 8               <ul style="list-style-type: none"> <li>▶ Leucovorin 15 mg PO (preferred) or IM every 12 hours for 4 doses beginning 24 hours after the start of the Methotrexate infusion</li> </ul> </li> <li>• Dactinomycin 0.5 mg IV Push on Day 8</li> <li>• Filgrastim 5 mcg/kg SC on Days 3–6 and 10–13<sup>c</sup></li> </ul>                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

<sup>a</sup> FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.

<sup>b</sup> Use G-CSF as primary prophylaxis with etoposide/cisplatin or etoposide/carboplatin-based regimens. See [NCCN Guidelines for Hematopoietic Growth Factors](#).

<sup>c</sup> Round filgrastim dose to the nearest vial size by institution-defined weight limits.

<sup>d</sup>Dosing schedules also apply to the Intermediate Trophoblastic Tumor (PSTT and ETT) regimens listed on [GTN-D 7 of 7](#).

**Continued**

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
All recommendations are category 2A unless otherwise indicated.

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# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### SYSTEMIC THERAPY FOR GTN

##### High-Risk GTN: Therapy for Methotrexate-Resistant GTN<sup>a,b</sup>

| Preferred                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TP-TE (Paclitaxel/Cisplatin-Paclitaxel/Etoposide) (Repeat every 2 weeks)<sup>d</sup></b> <ul style="list-style-type: none"><li>• Paclitaxel 135 mg/m<sup>2</sup> IV over 3 hours on Day 1</li><li>• Cisplatin 60–75 mg/m<sup>2</sup> IV on Day 1</li></ul> <b>Alternating every 2 weeks with:</b> <ul style="list-style-type: none"><li>• Paclitaxel 135 mg/m<sup>2</sup> IV over 3 hours on Day 15</li><li>• Etoposide 150 mg/m<sup>2</sup> IV over 60 minutes on Day 15</li><li>• Pegfilgrastim 6 mg SC on Days 2 and 16<br/>OR Filgrastim 5 mcg/kg on Days 2–6 and 16–20</li></ul>                                                                                                                                                                                                     | <b>VIP (Etoposide/Ifosfamide/Mesna/Cisplatin) (Repeat every 3 weeks)<sup>d,e</sup></b> <ul style="list-style-type: none"><li>• Etoposide 75 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1–5</li><li>• Ifosfamide 1200 mg/m<sup>2</sup>/day IV over 3 hours on Days 1–5 with mesna protection<ul style="list-style-type: none"><li>▶ Mesna 120 mg/m<sup>2</sup>/day IV bolus 15 minutes prior to Ifosfamide, then 1200 mg/m<sup>2</sup>/day IV infusion over 12 hours after Ifosfamide dose on Days 1–5</li></ul></li><li>• Cisplatin 20 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1–5</li><li>• Pegfilgrastim 6 mg SC on Day 6;<br/>OR Filgrastim 5 mcg/kg SC on Days 6–14<sup>c</sup></li></ul> |
| <b>BEP (Bleomycin/Etoposide/Cisplatin) (Repeat every 3 Weeks)<sup>d,e</sup></b> <ul style="list-style-type: none"><li>• Bleomycin 30 units IV Push weekly on Days 1, 8, and 15 or Days 2, 9, and 16</li><li>• Etoposide 100 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1–5</li><li>• Cisplatin 20 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1–5</li><li>• Pegfilgrastim 6 mg SC on Day 6;<br/>OR Filgrastim 5 mcg/kg SC on Days 10–14<sup>c</sup></li></ul> <b>Comments/Considerations</b> <ul style="list-style-type: none"><li>• Assessment of pulmonary function tests (PFTs), typically spirometry and diffusing capacity for carbon monoxide (DLCO), at baseline prior to treatment and monitoring PFTs prior to each new treatment cycle as clinically indicated.</li></ul> | <b>ICE (Ifosfamide/Mesna/Carboplatin/Etoposide) (Repeat every 3 Weeks)<sup>d</sup></b> <ul style="list-style-type: none"><li>• Ifosfamide 1.2 g/m<sup>2</sup>/day IV on Days 1–3 with mesna protection<ul style="list-style-type: none"><li>▶ Mesna 120 mg/m<sup>2</sup>/day IV bolus 15 minutes prior to Ifosfamide, then 1200 mg/m<sup>2</sup>/day IV infusion over 12 hours after Ifosfamide dose on Days 1–3</li></ul></li><li>• Carboplatin AUC 4 IV over 30 minutes on Day 1</li><li>• Etoposide 75 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1–3</li><li>• Pegfilgrastim 6 mg SC on Day 4;<br/>OR Filgrastim 5 mcg/kg SC on Days 4–14</li></ul>                                               |

<sup>a</sup> FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.

<sup>b</sup> Use G-CSF as primary prophylaxis with etoposide/cisplatin or etoposide/carboplatin-based regimens. See [NCCN Guidelines for Hematopoietic Growth Factors](#).

<sup>c</sup> Round filgrastim dose to the nearest vial size by institution-defined weight limits.

<sup>d</sup> Dosing schedules also apply to the Intermediate Trophoblastic Tumor (PSTT and ETT) regimens listed on [GTN-D 7 of 7](#).

<sup>e</sup> For dosing references, see page TEST-E in the [NCCN Guidelines for Testicular Cancer](#).

[Continued](#)

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
All recommendations are category 2A unless otherwise indicated.



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### SYSTEMIC THERAPY FOR GTN

##### High-Risk GTN: Therapy for Methotrexate-Resistant GTN<sup>a,b</sup>

###### Other Recommended

###### **TIP (Paclitaxel/Ifosfamide/Mesna/Cisplatin) (Repeat every 3 Weeks)<sup>f</sup>**

- Paclitaxel 250 mg/m<sup>2</sup> IV over 24 hours on Day 1
- Ifosfamide 1500 mg/m<sup>2</sup>/day IV over 3 hours on Days 2–5 with mesna protection
  - ▶ Mesna 300 mg/m<sup>2</sup> IV bolus 15 minutes prior to Ifosfamide, then at 4 hours and 8 hours from the start of each Ifosfamide dose on Days 2–5
- Cisplatin 25 mg/m<sup>2</sup>/day IV over 60 minutes on Days 2–5
- Pegfilgrastim 6 mg SC on Day 6 OR Filgrastim 5 mcg/kg SC on Days 6–14

<sup>a</sup> FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.

<sup>b</sup> Use G-CSF as primary prophylaxis with etoposide/cisplatin or etoposide/carboplatin-based regimens. See [NCCN Guidelines for Hematopoietic Growth Factors](#).

<sup>f</sup> For dosing references, see page TEST-F in the [NCCN Guidelines for Testicular Cancer](#).

**Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF**  
**All recommendations are category 2A unless otherwise indicated.**

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**GTN-D  
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# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### SYSTEMIC THERAPY FOR GTN

##### High-Risk GTN:

**Additional Agents/Regimens Shown to Have Some Activity in Treating Multiagent Chemotherapy-Resistant GTN<sup>d,9</sup>**

| Useful in Certain Circumstances                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• <b>PD-1/PD-L1 inhibitors<sup>h</sup></b><br/>(eg, <b>Pembrolizumab<sup>i</sup> [preferred]</b>, <b>Nivolumab<sup>j</sup> [preferred]</b>, <b>Avelumab<sup>h</sup></b>)<ul style="list-style-type: none"><li>▶ Pembrolizumab 200 mg IV over 30 minutes every 3 Weeks or 400 mg IV over 30 minutes every 6 Weeks</li><li>▶ Nivolumab 240 mg IV over 30 minutes every 2 Weeks or 480 mg IV over 30 minutes every 4 Weeks</li><li>▶ Avelumab 800 mg IV over 60 minutes every 2 Weeks</li></ul></li><li>• <b>Ipilimumab + Nivolumab<sup>h</sup></b><ul style="list-style-type: none"><li>▶ Nivolumab 240 mg IV every 2 Weeks</li><li>▶ Ipilimumab 1 mg/kg IV every 6 Weeks</li></ul></li><li>• <b>Capecitabine-based regimens<sup>k</sup> (Repeat cycle every 3 Weeks)</b><ul style="list-style-type: none"><li>▶ Capecitabine 1250 mg/m<sup>2</sup> PO twice daily for 2 Weeks on and 1 Week off</li></ul></li><li>• <b>Gemcitabine ± Cisplatin (Repeat every 4 Weeks)</b><ul style="list-style-type: none"><li>▶ Gemcitabine 600–800 mg/m<sup>2</sup>/day IV over 30 minutes on Days 1, 8, and 15</li><li>▶ Cisplatin 25–30 mg/m<sup>2</sup>/day IV over 60 minutes on Days 1, 8, and 15 (Repeat every 4 Weeks)</li></ul></li><li>• <b>High-dose chemotherapy with peripheral hematopoietic cell transplant (HCT)<sup>l</sup></b></li></ul> |

<sup>d</sup> Dosing schedules also apply to the Intermediate Trophoblastic Tumor (PSTT and ETT) regimens listed on [GTN-D 7 of 7](#).

<sup>9</sup> Recommend referral to center that specializes in the treatment of GTN.

<sup>h</sup> [NCCN Guidelines for the Management of Immune Checkpoint Inhibitor-Related Toxicities](#).

<sup>i</sup> Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.

<sup>j</sup> Nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.

<sup>k</sup> For information regarding dihydropyrimidine dehydrogenase (DPYD) testing, see the [NCCN Guidelines for Colon Cancer](#).

<sup>l</sup> Given on clinical trial protocol.

**Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).**  
**All recommendations are category 2A unless otherwise indicated.**

[Continued](#)



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### SYSTEMIC THERAPY FOR GTN

##### Intermediate Trophoblastic Tumor (PSTT and ETT)<sup>a,b,m,n</sup>

| Preferred                                                                                                                                                                                     | Other Recommended                                                                                                                                                                                                                                             | Useful in Certain Circumstances                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• EMA-EP (Etoposide/Methotrexate/Dactinomycin-Etoposide/Cisplatin)</li><li>• EP-EMA (Etoposide/Cisplatin-Etoposide/Methotrexate/Dactinomycin)</li></ul> | <ul style="list-style-type: none"><li>• TP-TE (Paclitaxel/Cisplatin-Paclitaxel/Etoposide)</li><li>• BEP (Bleomycin/Etoposide/Cisplatin)</li><li>• VIP (Etoposide/Ifosfamide/Mesna/Cisplatin)</li><li>• ICE (Ifosfamide/Mesna/Carboplatin/Etoposide)</li></ul> | <p><b><u>Additional agents/regimens shown to have some activity in treating multiagent chemotherapy-resistant GTN:</u></b></p> <ul style="list-style-type: none"><li>• PD-1/PD-L1 inhibitors (eg, Pembrolizumab,<sup>i</sup> Nivolumab,<sup>j</sup> Avelumab)<sup>h</sup></li><li>• Capecitabine-based regimens<sup>k</sup></li><li>• Gemcitabine ± Cisplatin</li><li>• High-Dose chemotherapy with peripheral HCT<sup>l</sup></li><li>• Ipilimumab + Nivolumab<sup>h</sup></li></ul> |

#### [See Evidence Blocks on EB-2](#)

<sup>a</sup> FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.

<sup>b</sup> Use G-CSF as primary prophylaxis with etoposide/cisplatin or etoposide/carboplatin-based regimens. See [NCCN Guidelines for Hematopoietic Growth Factors](#).

<sup>h</sup> [NCCN Guidelines for the Management of Immune Checkpoint Inhibitor-Related Toxicities](#).

<sup>i</sup> Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.

<sup>j</sup> Nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.

<sup>k</sup> For information regarding dihydropyrimidine dehydrogenase (DPYD) testing, see the [NCCN Guidelines for Colon Cancer](#).

<sup>l</sup> Given on clinical trial protocol.

<sup>m</sup> If feasible, perform hysterectomy with salpingectomy and excision of metastatic disease.

<sup>n</sup> For dosing schedules, see High-Risk GTN sections [GTN-D 3 of 7](#), [GTN-D 4 of 7](#), and [GTN-D 6 of 7](#).

**Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#). All recommendations are category 2A unless otherwise indicated.**



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### PRINCIPLES OF GYNECOLOGIC SURVIVORSHIP

##### Physical Effects

- Gynecologic cancer treatment typically involves surgery, chemotherapy, hormone therapy, radiation therapy (RT), and/or immunotherapy. These treatments cause acute, short-term, and long-term toxicities.
- Surgical approaches may be extensive and pose risks such as adhesion formation, which may cause pain and may contribute to small bowel obstruction, urinary or gastrointestinal complications (eg, incontinence, diarrhea), pelvic floor dysfunction (manifested by a variety of urinary, bowel, and/or sexual effects), and lymphedema.
- Chemotherapy agents vary, though commonly used regimens may pose a significant risk of neurotoxicity, cardiac toxicity, development of hematologic cancers, and cognitive dysfunction.
- Long-term estrogen deprivation may cause symptoms such as hot flashes, vaginal dryness, and bone loss.
- RT may cause long-term complications (eg, fibrosis, vulvovaginal atrophy) and may predispose patients to secondary cancers of the subcutaneous tissue, and/or underlying organs that are proximal to the radiation field.
- Prior pelvic RT may contribute to bone loss and increase the risk of pelvic fractures. Consider bone density testing and prophylactic use of bisphosphonates, particularly in patients with osteoporosis.
- Immunotherapy use is emerging, and to date, long-term effects of these treatments are unknown.

##### Psychosocial Effects

- Psychosocial effects after cancer may be psychological (eg, depression, anxiety, fear of recurrence, altered body image), financial (eg, return to work, insurance concerns), and/or interpersonal (eg, relationships, sexuality, intimacy) in nature.

##### Clinical Approach

- All gynecologic cancer survivors should receive regular general medical care that focuses on managing chronic disease, monitoring cardiovascular risk factors, providing recommended vaccinations, and encouraging adoption of a healthy lifestyle.
- In order to assess the late and long-term effects of gynecologic cancers, clinicians should comprehensively document the patient's history, conduct a thorough physical examination, and provide any necessary imaging and/or laboratory testing. All patients, whether sexually active or not, should be asked about genitourinary symptoms, including vulvovaginal dryness. Referral to appropriate specialty providers (eg, physical therapy, pelvic floor therapy, sexual therapy, psychotherapy) is recommended. As most treatments for gynecologic cancers will cause sexual dysfunction, early menopause, and infertility, special attention to the resultant medical and psychosocial implications is needed.
- Post-radiation use of vaginal dilators and moisturizers is recommended.
- For treatment-related menopause, hormone therapy should be considered.
- Communication and coordination with all clinicians involved in the care of survivors, including primary care clinicians, is critical. Providing cancer survivors with a summary of their treatment and recommendations for follow-up is recommended.
- In subsequent pregnancy:
  - ▶ Recommend early ultrasound to confirm intrauterine pregnancy
  - ▶ Perform pathologic evaluation of placenta
  - ▶ Check hCG at 6 weeks postpartum to confirm return to negative

##### Additional Guidance

- [NCCN Guidelines for Distress Management](#)
- [NCCN Guidelines for Smoking Cessation](#)
- [NCCN Guidelines for Survivorship](#)

**Note:** For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#).  
All recommendations are category 2A unless otherwise indicated.



|   |   |   |   |   |
|---|---|---|---|---|
| 4 |   |   |   |   |
| 3 |   |   |   |   |
| 2 |   |   |   |   |
| 1 |   |   |   |   |
|   | E | S | Q | C |

**E** = Efficacy of Regimen/Agent  
**S** = Safety of Regimen/Agent  
**Q** = Quality of Evidence  
**C** = Consistency of Evidence  
**A** = Affordability of Regimen/Agent

## EVIDENCE BLOCKS FOR PRINCIPLES OF SYSTEMIC THERAPY

| REGIMENS FOR LOW-RISK GTN |  |
|---------------------------|--|
| Preferred Regimens        |  |
| Methotrexate              |  |
| Dactinomycin              |  |

| HIGH-RISK GTN: PRIMARY THERAPY OPTIONS                                       |  |
|------------------------------------------------------------------------------|--|
| Preferred Regimens                                                           |  |
| EMA/CO (Etoposide, Methotrexate, Dactinomycin/Cyclophosphamide, Vincristine) |  |
| EMA/EP (Etoposide, Methotrexate, Dactinomycin/Etoposide, Cisplatin)          |  |
| EP/EMA (Etoposide, Cisplatin/Etoposide, Methotrexate, Dactinomycin)          |  |

| HIGH-RISK GTN: THERAPY FOR METHOTREXATE-RESISTANT GTN |  |
|-------------------------------------------------------|--|
| Preferred Regimens                                    |  |
| TP/TE (Paclitaxel, Cisplatin/Paclitaxel, Etoposide)   |  |
| BEP (Bleomycin, Etoposide, Cisplatin)                 |  |
| VIP (Etoposide, Ifosfamide, Cisplatin)                |  |
| ICE (Ifosfamide, Carboplatin, Etoposide)              |  |
| Other Recommended Regimens                            |  |
| TIP (Paclitaxel, Ifosfamide, Cisplatin)               |  |
| Useful in Certain Circumstances                       |  |
| Pembrolizumab                                         |  |
| Nivolumab                                             |  |
| Avelumab                                              |  |
| Nivolumab/Ipilimumab                                  |  |
| Capecitabine                                          |  |
| Gemcitabine                                           |  |
| Gemcitabine/Cisplatin                                 |  |

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#)



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

|   |   |   |   |   |   |
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|   | E | S | Q | C | A |

**E** = Efficacy of Regimen/Agent  
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**A** = Affordability of Regimen/Agent

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#### EVIDENCE BLOCKS FOR PRINCIPLES OF SYSTEMIC THERAPY

| INTERMEDIATE TROPHOBLASTIC TUMOR (PSTT AND ETT)                     |  |
|---------------------------------------------------------------------|--|
| Preferred Regimens                                                  |  |
| EMA/EP (Etoposide, Methotrexate, Dactinomycin/Etoposide, Cisplatin) |  |
| EP/EMA (Etoposide, Cisplatin/Etoposide, Methotrexate, Dactinomycin) |  |
| Other Recommended Regimens                                          |  |
| TP/TE (Paclitaxel, Cisplatin/Paclitaxel, Etoposide)                 |  |
| BEP (Bleomycin, Etoposide, Cisplatin)                               |  |
| VIP (Etoposide, Ifosfamide, Cisplatin)                              |  |
| ICE (Ifosfamide, Carboplatin, Etoposide)                            |  |
| Useful in Certain Circumstances                                     |  |
| Pembrolizumab                                                       |  |
| Nivolumab                                                           |  |
| Avelumab                                                            |  |
| Capecitabine                                                        |  |
| Gemcitabine                                                         |  |
| Gemcitabine/Cisplatin                                               |  |
| Nivolumab/Ipilimumab                                                |  |

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page [EB-DEF](#)



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## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

**FIGO STAGING SYSTEM FOR GTN<sup>a</sup>**

| Stage | Criteria                                                                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------|
| I     | Tumor confined to uterus                                                                                           |
| II    | Tumor extends to other genital structures (ovary, tube, vagina, broad ligaments) by metastasis or direct extension |
| III   | Lung metastasis                                                                                                    |
| IV    | All other distant metastases                                                                                       |

<sup>a</sup>Used with permission of the American College of Surgeons, Chicago, Illinois. The original source for this information is the AJCC Cancer Staging Manual, Eighth Edition (2017) published by Springer International Publishing.



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## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

#### PROGNOSTIC SCORING INDEX FOR GTN<sup>a</sup>

| Prognostic factor                         | Risk score        |                                     |                                    |                   |
|-------------------------------------------|-------------------|-------------------------------------|------------------------------------|-------------------|
|                                           | 0                 | 1                                   | 2                                  | 4                 |
| Age (years)                               | <40               | ≥40                                 | —                                  | —                 |
| Antecedent pregnancy                      | Hydatidiform mole | Abortion                            | Term pregnancy                     | —                 |
| Interval from index pregnancy (months)    | <4                | 4-6                                 | 7-12                               | >12               |
| Pretreatment hCG (IU/L)                   | <10 <sup>3</sup>  | 10 <sup>3</sup> to <10 <sup>4</sup> | 10 <sup>4</sup> to 10 <sup>5</sup> | ≥10 <sup>5</sup>  |
| Largest tumor size, including uterus (cm) | <3                | 3-5                                 | >5                                 |                   |
| Site of metastases                        | Lung              | Spleen, kidney                      | Gastrointestinal tract             | Brain, liver      |
| Number of metastases identified           | 0                 | 1-4                                 | 5-8                                | >8                |
| Previous failed chemotherapy              | —                 | —                                   | Single drug                        | Two or more drugs |
| Total score                               | —                 | —                                   | —                                  | —                 |

- The total score for a patient is obtained by adding the individual scores for each prognostic factor.
- FIGO Prognostic Score
  - ▶ Low risk: <7
  - ▶ High risk: ≥7

<sup>a</sup>Used with permission of the American College of Surgeons, Chicago, Illinois. The original source for this information is the AJCC Cancer Staging Manual, Eighth Edition (2017) published by Springer International Publishing.





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## Gestational Trophoblastic Neoplasia

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# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

### NCCN Evidence Blocks™

| NCCN Categories of Evidence and Consensus |                                                                                                                                                                                                        |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Category 1</b>                         | Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate. |
| <b>Category 2A</b>                        | Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.                                                                     |
| <b>Category 2B</b>                        | Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.                                                                   |
| <b>Category 3</b>                         | Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.                                                                                               |

All recommendations are category 2A unless otherwise indicated.

| NCCN Categories of Preference          |                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preferred</b>                       | Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.                                              |
| <b>Other recommended</b>               | Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes. |
| <b>Useful in certain circumstances</b> | Other interventions that may be used for selected patient populations (defined with recommendation).                                                        |

All recommendations are considered appropriate.



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## Gestational Trophoblastic Neoplasia

### Discussion

This discussion corresponds to the NCCN Guidelines for Gestational Trophoblastic Neoplasia. Last updated on 11/21/2025

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## Gestational Trophoblastic Neoplasia

### Overview

Gestational trophoblastic disease (GTD) refers to a group of benign and malignant tumors that develop in the uterus from placental tissue. The pathogenesis of GTD is unique in that maternal tumors arise from gestational tissue that can have locally invasive or metastatic potential. Historical data on incidence of GTD varies widely by region, with higher incidence reported in Asia compared with Europe and North America. These differences are thought to be due at least in part to varying diagnostic criteria, reporting practices, quality of epidemiologic data, and diet and nutrition. In the United States, the reported incidence of GTD is approximately 1 out of every 1000 pregnancies.<sup>1-4</sup>

The most common form of GTD is hydatidiform mole (HM), also known as molar pregnancy. HMs are considered a benign, premalignant disease. Malignant forms of GTD are collectively referred to as gestational trophoblastic neoplasia (GTN), and include invasive mole, choriocarcinoma, placental site trophoblastic tumor (PSTT), and epithelioid trophoblastic tumor (ETT). HM encompasses about 80% of all GTD, invasive moles account for 15%, and choriocarcinoma and other rarer types of GTN comprise the remaining 5%.<sup>5</sup> Cure rates are approaching 100%, and treatment typically allows for fertility preservation.<sup>5,6</sup>

### Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at [www.NCCN.org](http://www.NCCN.org).

### Literature Search Criteria

Prior to the update of this version of the NCCN Clinical Practice Guidelines (NCCN Guidelines®) for Gestational Trophoblastic Neoplasia, an electronic search of the PubMed database was performed to obtain key literature in GTD published since the previous Guidelines update, using the following search terms: gestational trophoblastic OR choriocarcinoma OR

intermediate trophoblastic tumor. The PubMed database was chosen as it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase II; Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies.

The data from key PubMed articles as well as articles from additional sources deemed as relevant to these Guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

### Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation. NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-fat-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs



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## Gestational Trophoblastic Neoplasia

present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

### Types of Gestational Trophoblastic Disease

HM occurs as a result of abnormal fertilization and is characterized as complete or partial based on differences in morphology, karyotype, and malignant potential. The majority of complete moles (80%) occur as a result of abnormal fertilization of an ovum lacking nuclear DNA and have two identical paternal chromosome complements derived from duplication of the haploid genome of a single sperm. The remaining 20% occur as a result of dispermy (fertilization by two sperms). Partial moles occur when an ovum retains its nucleus and abnormal fertilization occurs in one of two ways: 1) fertilization by a single sperm with subsequent paternal chromosome duplication; or 2) via dispermy. Partial HMs can contain fetal tissue, but complete moles do not. More information on the pathologic diagnosis of benign, noninvasive HM can be found in the *Principles of Pathology* section in the algorithm.

Post-molar GTN, which includes invasive mole and choriocarcinoma, develops in about 15% to 20% of complete moles, but in only 1% to 5% of partial moles.<sup>2,3,7,8</sup> The reported incidence of GTN after molar pregnancy is 18% to 29%.<sup>2,3,9,10</sup> This rate appears to be stable despite the progressively earlier diagnosis of complete HM.<sup>10</sup> Invasive moles arise from extension of HM into the myometrium via tissue or venous channels. Approximately 15% of invasive moles metastasize to the lung or vagina. Persistent elevated human chorionic gonadotropin (hCG) after evacuation of a molar pregnancy most often leads to the diagnosis of invasive mole.<sup>2</sup> Choriocarcinoma develops from villous trophoblast. Features of these malignant epithelial tumors include abnormal trophoblastic hyperplasia and anaplasia, hCG production, absence of chorionic villi, hemorrhage,

and necrosis.<sup>2,3</sup> Choriocarcinoma has been reported to occur with different types of pregnancy events, including HM (50%), term or preterm gestation (25%), and tubal pregnancy or abortion (25%). Approximately 2% to 3% of HMs progress to choriocarcinoma.

Intermediate trophoblastic tumors (ITT), including PSTT and ETT, are rare subtypes of GTN with an incidence of about 1 in 100,000 pregnancies, representing approximately 1% of all GTN cases.<sup>11</sup> Most PSTTs follow nonmolar gestations and present months to years after the antecedent pregnancy. Less often, PSTT develops after evacuation of HM.<sup>5</sup> PSTT arises from interstitial trophoblast at the placental implantation site and consists predominately of mononuclear intermediate trophoblast without chorionic villi, infiltrating in sheets or cords between myometrial fibers. It is associated with less vascular invasion, necrosis, and hemorrhage than choriocarcinoma.

ETT is a rare variant of PSTT that simulates carcinoma. Based on morphologic and histochemical features, it appears to develop from neoplastic transformation of chorionic-type intermediate trophoblast. ETT typically presents years after term delivery.

### Hydatidiform Mole

#### Presentation and Workup

Patients with HM commonly present with vaginal bleeding, typically around 6 to 16 weeks of gestation. Due to widespread ultrasound screening during early pregnancy and accurate hCG testing, most cases of HM are detected prior to the onset of additional signs such as uterine enlargement beyond that expected for gestation date, preeclampsia, hyperemesis, anemia, and theca lutein ovarian cysts.<sup>2,3,5</sup> Partial HMs tend to grow more slowly and may present later in the first or early second trimester, often with symptoms of incomplete or missed abortion and diagnosis made upon histologic examination of the curettage specimen.<sup>2,3</sup>



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## Gestational Trophoblastic Neoplasia

Initial determination of suspected HM is often made based on ultrasound findings in combination with clinical symptoms and hCG levels. Due to hyperplastic trophoblastic cells in complete HM, many patients will have marked elevations in hCG, at times >100,000 IU/L. However, such elevations in hCG are observed in <10% of patients with partial HM. Characteristic ultrasound findings of complete HM include enlarged uterus with a heterogenous mass (ie, snowstorm appearance). Hydropic/swollen chorionic villi lead to the appearance of small cystic spaces, creating a vesicular pattern. However, these characteristics may not be readily observed with the diagnosis of HM early in the first trimester. As molar pregnancy advances, these cystic spaces become larger and more numerous. Features that may be noted on ultrasound imaging of partial HM include focal cystic spaces within the placenta, gestational sac that is empty or elongated along the transverse axis, and/or fetal anomalies or fetal demise.<sup>2,3,5,9,12</sup> The NCCN Panel recommends workup of patients with HM to include history and physical; pelvic ultrasound; quantitative hCG assay; complete blood count (CBC) with platelet count; liver, renal, and thyroid function tests; as well as blood type and screen (administer Rho[D] immune globulin if Rh-negative). Recommended imaging also includes chest x-ray. If chest x-ray is positive for metastases, perform chest/abdomen/pelvis (C/A/P) CT with contrast and brain MRI with or without contrast; manage as GTN after initial uterine evacuation. Distress should be assessed and treated promptly at all stages of disease and in all settings. The NCCN Distress thermometer for screening distress can be found in NCCN Guidelines for Distress Management at [www.NCCN.org](http://www.NCCN.org).

### Treatment

Initial treatment of HM in patients who wish to preserve fertility is suction dilation and curettage (D&C), preferably performed under ultrasound guidance to reduce the risk of uterine perforation.<sup>9,13</sup> To reduce the risk of heavy bleeding, uterotonic agents (eg, methylergonovine and/or prostaglandins) should be administered during the procedure and

continued for several hours postoperatively.<sup>2,14</sup> Tranexamic acid (TXA) can be considered if necessary. For patients who do not wish to preserve fertility, hysterectomy with salpingectomy can be considered as an alternative.<sup>15</sup> Histopathologic review and DNA genotyping help confirm the diagnosis of benign, noninvasive HM,<sup>16</sup> and may allow patients at low risk for developing post-molar GTN (ie, those with partial HM) to avoid unnecessary treatment with prophylactic chemotherapy.<sup>17</sup>

Prophylactic chemotherapy at the time of uterine evacuation is controversial and may reduce the incidence of post-molar GTN by 3% to 8%. A Cochrane database review (three randomized controlled trials [RCTs], n = 613) did not conclude sufficient evidence for standard administration of prophylactic chemotherapy to prevent post-molar GTN; however, evidence was suggestive that prophylactic chemotherapy may reduce the risk of progression to GTN among patients with complete HM at high risk for malignant transformation.<sup>18</sup> The NCCN Guidelines state that prophylactic methotrexate or dactinomycin can be considered for patients deemed at high risk for post-molar GTN (eg, age >40 years, hCG levels in excess of 100,000 mIU/mL, excessive uterine enlargement, and/or theca lutein cysts >6 cm).<sup>2,9,18,19</sup>

### Follow-up

Follow-up with hCG monitoring is essential following initial treatment of HM to ensure that hCG levels return to normal. The hCG molecules associated with GTD are more heterogenous and degraded than those associated with normal pregnancy.<sup>2,20</sup> Therefore, monitoring should be performed with a quantitative assay capable of detecting all forms of hCG, including beta-hCG, core hCG, nicked-free beta, beta core, and hyperglycosylated forms.<sup>5,21,22</sup> Post-molar GTN develops in about 15% to 20% of complete moles, but in only 0.1% to 5% of partial moles. Therefore, careful monitoring can facilitate early detection of persistent GTN. Risk of recurrence is low (<2%) following a single molar pregnancy,





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but increases significantly for patients who experience one or more recurrences.<sup>2,3,7,13,14,23</sup>

Once normalized, recurrent elevation of hCG has been reported in <1% of patients.<sup>23,24</sup> The occurrence of GTN following hCG normalization is rare after the recommended 6 months of post-normalization hCG monitoring.<sup>25</sup> A 2018 study showed that patients with complete HM who normalized beyond 56 days post uterine evacuation had a 3.8-fold higher risk of developing post-molar GTN.<sup>23</sup> In another retrospective population study,<sup>26</sup> 17,424 patients with hCG normalization after evacuation of complete HM were included to provide a reduced surveillance protocol to detect GTN post hCG normalization. The risk of developing gestational trophoblastic neoplasia after uterine evacuation was substantially lower if hCG levels returned to normal in <56 days rather than ≥56 days (posterior medians, 0.06%; 95% credible interval, 0.01%–0.14% vs. 0.22%, 0.15%–0.31%), with a posterior relative risk of 0.25 (0.06–0.72). Most patients who developed gestational trophoblastic neoplasia (71.0%, n = 22 of 31) had received a diagnosis after the current 6-month surveillance protocol. This study recommends a change in the surveillance protocols to one confirmatory normal hCG value for patients whose hCG levels return to normal in <56 days of evacuation of a complete HM. For patients whose hCG levels return to normal in ≥56 days, counseling is required to decide the length of subsequent follow-up considering the risk of GTN over time.<sup>26</sup>

The NCCN Panel recommends hCG assay monitoring every 1 to 2 weeks until levels have normalized, defined in the guidelines as three consecutive normal assays. Patients should also use contraception; oral contraceptive pills are preferred over intrauterine devices because they suppress endogenous luteinizing hormone (LH)/follicle-stimulating hormone (FSH), which may interfere with hCG measurement at low levels. Following initial normalization after 3 consecutive assays, one additional hCG assay at least 1 month from the final consecutively normal hCG

assay is required for partial HM.<sup>24,27,28</sup> For complete HM, monthly hCG assay for 3 to 6 months is advised to ensure levels remain normal. If time from D&C to hCG normalization is <56 days, 3 months can be considered and if it is ≥56 days, monthly hCG assays can extend up to 6 months.<sup>29</sup> If hCG levels remain persistently elevated (ie, plateau or rise), treat per the persistent post-molar GTN algorithm.

### Post-Molar GTN

Post-molar GTN is typically diagnosed by hCG surveillance. The NCCN Guidelines use the FIGO staging criteria for post-molar GTN as meeting one of more of the following criteria after treatment for HM, as indicated by hCG monitoring<sup>30</sup> and WHO prognostic scoring index for GTN<sup>31</sup>:

- hCG levels plateau for 4 consecutive values over 3 weeks
- hCG levels rise ≥10% for 3 values over 2 weeks

Assessment and staging of post-molar GTN should include history and physical examination (including pelvic exam), FIGO stage and prognostic score, pelvic ultrasound, and imaging (chest x-ray or [C/A/P] CT to assess for metastatic disease. Pelvic ultrasound is used to confirm the absence of pregnancy, measure uterine size, and to delineate the volume and vasculature of the tumor. If chest x-ray reveals no evidence of metastatic disease, no further imaging is recommended prior to treatment. If the chest x-ray suggests metastases, C/A/P CT scan with contrast and brain MRI with or without are indicated. Post-treatment imaging is not routinely recommended for follow-up after hCG normalization in patients with post-molar GTN or choriocarcinoma, where hCG is a reliable tumor marker. It may be necessary to follow patients with brain metastases with MRI brain with contrast for 2 to 3 years.

Repeat D&C for all patients with WHO ≤4, hysterectomy with salpingectomy, or single-agent systemic therapy can be considered for persistent post-molar GTN with no evidence of extrauterine disease.<sup>32-34</sup>





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An observational study conducted over a period of 10 years examined 544 patients who underwent second uterine evacuation for persistent GTD.<sup>34</sup> Following repeat curettage, 68% had no further evidence of disease or chemotherapy requirements. However, chemotherapy requirement was more likely for patients with a histologic confirmation of persistent trophoblastic disease and for urinary hCG levels in excess of 1500 IU/L at time of second evacuation.<sup>34</sup> Several groups have discussed the optimal characteristics of candidates for repeat uterine evacuation.<sup>34-38</sup>

Hysterectomy with salpingectomy or single-agent systemic therapy should be followed by hCG monitoring every 2 weeks until the patient has 3 consecutive normal assays, followed by monthly hCG monitoring for 6 to 12 months. For persistent hCG elevation (ie, plateau or rise), follow GTN chemotherapy recommendations, which are stratified by patient response to initial therapy and the magnitude of hCG elevation.

For presence of metastatic disease, histopathologic diagnosis of choriocarcinoma, PSTT/ETT or extrauterine disease, follow recommendations for staging and treatment in the algorithms for GTN, which are stratified by risk group/prognostic score.

### Gestational Trophoblastic Neoplasia

#### Presentation and Workup

The presentation of GTN can vary depending upon the antecedent pregnancy event and disease type and extent. Post-molar GTN, including invasive mole or choriocarcinoma, can be associated with irregular bleeding after initial treatment for molar pregnancy, an enlarged and irregular uterus, and bilateral ovarian enlargement. However, these signs may be absent in patients with choriocarcinoma associated with normal, non-molar pregnancies. Trophoblastic tumors have fragile vessels and as a result, metastatic lesions are often hemorrhagic. In addition to bleeding, metastatic lesions may be associated with neurologic or pulmonary

symptoms. ETT and PSTT typically present with irregular uterine bleeding arising after some time has passed from a previous pregnancy.<sup>2,3,39</sup>

Workup for GTN includes history and physical examination and metastatic imaging workup, to include C/A/P CT scan with contrast (or MRI if contrast is contraindicated) as well as brain MRI (preferred) with and without contrast or CT with contrast if pulmonary metastasis. Pelvic ultrasound or MRI with or without contrast should also be performed. Visible lesions in the lower genital tract should not be biopsied due to hemorrhage risk. Additionally, the NCCN Panel recommends repeat CBC differential with platelets; liver, renal, and thyroid function testing; and hCG assay. If hCG is elevated with no evidence of disease on imaging, consider the possibility of LH crossover or phantom hCG.<sup>40,41</sup> If hCG is elevated, although at a low level that is stable over time (and false positive hCG ruled out), quiescent GTN is possible, and close follow-up, rather than immediate treatment may be considered. Based on these findings, GTN should be staged and scored according to the current FIGO staging and prognostic scoring system.<sup>30,42</sup> GTN staging is based on tumor location and extent: stage I disease is uterine-confined, stage II involves direct extension or metastasis to other genital structures, stage III disease is determined by lung metastasis, and stage IV disease includes non-pulmonary distant metastasis. The current FIGO prognostic scoring system was adapted from the WHO classification, which incorporated prognostic factors from Bagshawe's scoring system.<sup>43,44</sup> FIGO prognostic scoring is based on individual risk factors that have been shown to be predictive of GTN that is resistant to single-agent chemotherapy, such as age, antecedent pregnancy, interval from index pregnancy, pretreatment hCG, largest tumor size (including the uterus), site and number of metastases, and previous chemotherapy regimens that were unsuccessful. The sum of individual scores denotes the FIGO prognostic score of low-risk GTN (<7) or high-risk GTN (≥7).<sup>30,42,45</sup> This prognostic scoring system is not valid for the ITTs ETT and PSTT.<sup>11</sup>



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## Gestational Trophoblastic Neoplasia

Histopathologic assessment is also recommended by the Panel as part of initial workup. The immunohistochemical (IHC) markers melanoma cell adhesion molecule (Mel-CAM/CD146), human placental lactogen (hPL), beta-human chorionic gonadotropin ( $\beta$ -hCG), p63, cyclin E, and Ki-67 are all differentially expressed between PSTT, ETT, and choriocarcinoma.<sup>16,46,47</sup> GATA-3 is a sensitive marker for both benign and malignant trophoblast proliferations, and may be useful in distinguishing GTN from non-GTN tumors. Given that recommended treatment regimens for each of these GTN subtypes differ significantly, accurate diagnosis is essential for avoiding unnecessary treatment and optimizing patient outcomes.

### Low-Risk GTN

#### *First-Line Therapy*

As noted above, low-risk GTN is defined as a FIGO prognostic score of  $<7$ . Standard first-line treatment for low-risk GTN is single-agent chemotherapy using methotrexate or dactinomycin. Hysterectomy is preferred when fertility preservation is not desired. Ovaries can be left in situ even in the presence of theca lutein cysts. A retrospective French study<sup>48</sup> described the survival, efficacy and tolerance associated with first line hysterectomy in patients with low-risk GTN. In 74 patients with low-risk GTN who underwent hysterectomy, 61 patients observed hCG normalization (82.4%, 95% CI, 71.8–90.3), while 13 patients required follow-up salvage chemotherapy. Hysterectomy as a first-line treatment was reported effective in 82.4% of women with low-risk non-metastatic GTN. A second curettage is also recommended as an alternative for localized uterine disease. Hysterectomy without further systemic therapy is also recommended for patients without choriocarcinoma and a WHO score of  $<5$ . Numerous studies have evaluated these agents, but differences in inclusion criteria and dosage regimens have made it challenging to determine a superior regimen. While some consider methotrexate to have a more favorable adverse effect profile, dactinomycin may achieve similar

or better efficacy with a less-frequent infusion schedule.<sup>5,21,45,49,50</sup> A 2016 Cochrane Database review of RCTs in low-risk GTN showed with moderate-certainty evidence that first-line methotrexate may be more likely to fail than dactinomycin (risk ratio [RR], 3.55; 95% CI, 1.81–6.95; 6 trials, 577 participants;  $I(2) = 61\%$ ).<sup>50</sup> Similarly, the authors concluded that dactinomycin is more likely to lead to a primary cure than methotrexate (RR, 0.65; 95% CI, 0.57–0.75; 6 trials, 577 participants;  $I(2) = 26\%$ ).<sup>50</sup> However, 55% of the data came from trials of weekly IM methotrexate, which seems to be less effective than the 5- or 8-day methotrexate regimens. A closed phase III RCT ([NCT01535053](#)) comparing pulse dactinomycin to multiday methotrexate regimens noted primary remission rates of 75% for pulse dactinomycin versus 88.5% for the multiday methotrexate regimens (5-day  $>$  8-day). Overall quality-of-life scores were similar. Alopecia was more common with dactinomycin, mucositis was more common with the methotrexate regimens, and no patient required multiagent chemotherapy or salvage surgery to reach remission.<sup>51</sup>

Currently supported regimens of dactinomycin include a 5-day regimen (10–12 mcg/kg or flat 0.5 mg dose IV push, repeated every 2 weeks) or a dactinomycin pulse regimen (1.25 mg/m<sup>2</sup>, IV push, repeated every 2 weeks).<sup>21</sup> Primary remission rates for initial treatment with 5-day dactinomycin range from 77% to 94%, and for pulse dactinomycin, from 69% to 90%.<sup>45</sup> For methotrexate, currently supported regimens include 5-day methotrexate (0.4 mg/kg IV push or IM daily  $\times$  5 days, repeated every 2 weeks) or an 8-day regimen of methotrexate alternating with leucovorin rescue (1.0–1.5 mg/kg IM, every other day  $\times$  4 days, alternating with leucovorin, 15 mg PO, repeated every 2 weeks).<sup>21</sup> Primary remission rates for multiday methotrexate regimens range from 87% to 93% for the 5-day protocol, and from 74% to 93% for 8-day methotrexate with leucovorin rescue.<sup>45</sup> Both methotrexate and dactinomycin are category 1 preferred regimens for low-risk GTN.



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Methotrexate regimens that are no longer recommended due to lesser efficacy include weekly IM methotrexate (30–50 mg/m<sup>2</sup>) and pulse-dose IV infusion methotrexate.<sup>45,52,53</sup> Although weekly IM methotrexate was successful in 70% of patients with a prognostic score of 0–1, the success rate fell to 40% and 12% with a prognostic score of 2–4 and 5–6, respectively.<sup>5,52</sup> In a large case series (n = 618), 8-day methotrexate was comparatively more successful when analyzed by prognostic score subgrouping.<sup>17</sup>

The Guidelines note that a multiday methotrexate regimen is typically used as first-line therapy in low-risk GTN due to its generally favorable toxicity profile. Dactinomycin is often used as a secondary therapy for patients with methotrexate toxicity or effusions contradicting the use of methotrexate. Alternative single-agent options for treatment of low-risk GTN that are primarily used in Asia include etoposide and fluorouracil.<sup>45,54,55</sup>

### *Monitoring/Response Assessment During First-Line Therapy*

The NCCN Panel recommends monitoring chemotherapy response by hCG assay every 2 weeks, at the start of each treatment cycle.<sup>49</sup> Upon hCG normalization, continuation of therapy is recommended for two to three additional treatment cycles past normalization to minimize the risk of recurrence.<sup>3,6,21,56</sup> Surveillance should include monthly hCG for 1 year, along with contraception (oral contraception preferred). Chemotherapy resistance is indicated by a plateau in hCG (<10% change) over three consecutive cycles or a rise in hCG (>10% change) over two consecutive cycles.<sup>5,49</sup> Second-line chemotherapy is then indicated.

After initial therapy response, if hCG levels plateau or re-elevated (hCG <1000), workup needs to be repeated to check for metastasis to recalculate WHO score based on imaging. Second-line single agents that were not previously used can be used. Hysterectomy with salpingectomy can also be considered.

### *Second-Line Single-Agent Therapy*

Currently, there are no RCT data on second-line therapy for low-risk GTN, but general evidence and consensus supports a change to the alternative single-agent chemotherapy for patients who have had a good initial response to chemotherapy but experience hCG plateau or re-elevation (hCG <1000), or for patients who experience toxicity that limits the dose or frequency of treatment.<sup>5,21,57</sup> Increasing the hCG cut-off from 300 to 1000 IU/l for choosing patients for starting Actinomycin D following methotrexate resistance has been demonstrated to spare more patients from EMA/CO toxicity without compromising 100% survival outcomes.<sup>58</sup> Adjuvant hysterectomy with salpingectomy can be considered for patients with localized disease in the uterus for whom fertility preservation is not desired. The ovaries are left in situ, even in the presence of theca lutein cysts.

Second-line dactinomycin is considered to have an acceptable response rate in patients with low levels of hCG, but multi-agent chemotherapy may be favored in the second-line setting for patients whose hCG exceeds a given threshold.<sup>17,59,60</sup> The hCG threshold for considering dactinomycin versus multiagent regimens has been debated and revised over time.<sup>3,17,21,60,61</sup> The NCCN Guidelines note this hCG threshold as 1000 IU/l.

Dactinomycin has been associated with a complete response rate of approximately 75% in large case series of patients with methotrexate-resistant GTN.<sup>62,63</sup> A retrospective review of 358 patients with low-risk GTN identified 68 patients who were determined to have resistant disease after a 5-day methotrexate regimen (n = 68). The complete response rate to secondary dactinomycin was 75%, and all patients who required third-line multiagent chemotherapy with or without surgery achieved permanent remission. Clinicopathologic diagnosis of choriocarcinoma (vs. post-molar GTN) was significantly associated with resistance to secondary dactinomycin.<sup>62</sup> In a retrospective review of 877 patients with GTN initially treated with 8-day methotrexate, 103 patients required second-line therapy



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and were placed on a 5-day dactinomycin protocol.<sup>63</sup> Complete response to second-line dactinomycin was observed among 75.7% (n = 78). Among the 25 patients who required third-line treatment for resistant disease or relapse, overall survival was 100%.<sup>63</sup> The Guidelines also note that dactinomycin pulse regimen should not be used as secondary therapy for methotrexate-resistant disease nor as primary therapy in patients with choriocarcinoma.

### **Second-Line Multiagent Therapy**

For patients with poor response to single-agent chemotherapy or those who initially responded to initial therapy but experienced a subsequent rapid rise in hCG levels (hCG  $\geq 1000$ ) or exhibit poor response to initial therapy, repeat disease workup for metastasis, recalculate WHO score based on imaging and transition to combination chemotherapy.<sup>3,6</sup> Hysterectomy with salpingectomy can also be considered. The most commonly used regimen in this setting is EMA/CO (etoposide, methotrexate, and dactinomycin alternating with cyclophosphamide and vincristine).<sup>17,57,64</sup> The use of EMA/CO in this setting is based upon its efficacy in managing high-risk GTN.<sup>65</sup> Cure rates with EMA/CO approach 100% even in the presence of relapsed/resistant low-risk GTN.<sup>3,6,64</sup> As with single-agent second-line therapy, hysterectomy with salpingectomy may also be considered.

### **Monitoring/Response Assessment During Second-Line Therapy**

For both single-agent and multiagent second-line therapy, hCG levels should be monitored every 2 weeks, but a more rapid response assessment is warranted if hCG levels plateau or rise. The Panel recommends additional treatment if patients experience an hCG plateau over two consecutive treatment cycles or an hCG rise over 1 cycle. For persistent disease after single-agent therapy with or without hysterectomy/salpingectomy, repeat workup to assess for metastasis and transition to EMA/CO combination therapy. For persistent or recurrent disease after EMA/CO combination therapy with or without

hysterectomy/salpingectomy, treat per the high-risk GTN algorithm with platinum-based regimens and surgical resection as feasible.

### **High-Risk GTN**

High-risk GTN is defined as a prognostic score  $\geq 7$  or FIGO stage IV disease.<sup>30,42</sup> High-risk disease is relatively rare among patients with post-molar GTN, estimated at only 6% (39/618) in a large case series.<sup>17</sup> High-risk GTN should be treated with multiagent chemotherapy. Adjuvant surgery or radiation therapy may be included. With a multimodal approach, cure rates have reached approximately 90%, including almost all patients with only lung/vaginal metastases and 70% for patients with stage IV disease.<sup>6</sup> Factors associated with poorer outcomes include liver and brain metastases, particularly if co-occurring. However, the prognosis for these patients has improved over time.<sup>66-68</sup>

### **Primary Chemotherapy**

EMA/CO, in which EMA and CO are given on alternate weeks, is the most commonly used initial regimen for high-risk disease. Based on existing evidence, this regimen is thought to provide the best combination of efficacy with acceptable toxicity for treating patients with high-risk GTN. Multiple groups have confirmed the efficacy of EMA/CO, reporting complete response rates of 62% to 78% and long-term survival rates of 85% to 94%.<sup>64,65,69-76</sup>

Reports of other regimens that have been used in first-line treatment of high-risk GTN include EMA/EP (etoposide, methotrexate, dactinomycin alternating with etoposide, and cisplatin)<sup>77,78</sup> and EP/EMA (etoposide and cisplatin alternating with etoposide, methotrexate, and dactinomycin).<sup>79</sup>

Due to the lack of RCTs in this setting, systematic reviews have been unable to draw conclusions regarding a superior combination regimen for primary treatment of high-risk GTN.<sup>57,80</sup> EMA/EP (or EP/EMA) is highly active and considered by some to be superior to EMA/CO for ultra-high-





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risk disease; however, its use as standard initial therapy is limited by increased toxicity and inability to provide adequate salvage chemotherapy if required for persistent/recurrent disease.<sup>5,79</sup>

### *Induction Chemotherapy for Extensive Metastatic Disease*

Patients with widespread metastatic GTN, as evidenced by prognostic score >12, have a poorer prognosis.<sup>81,82</sup> Initiation of standard combination chemotherapy in these patients can lead to tumor collapse with hemorrhage, metabolic acidosis, septicemia, and/or multiple organ failure, resulting in the potential for early death (ie, within 4 weeks).<sup>21,64,82</sup> Efforts to improve outcomes for this ultra-high-risk population have included induction chemotherapy with etoposide and cisplatin prior to initiating EMA/CO.<sup>64,82</sup> In a case series of 140 patients with high-risk GTN, 33 patients who were determined to have large disease burden (ie, ultra-high-risk GTN) received low-dose induction chemotherapy with etoposide/cisplatin prior to EMA/CO therapy (etoposide 100 mg/m<sup>2</sup> IV and cisplatin 20 mg/m<sup>2</sup> IV on days 1 and 2, every 7 days for 1–3 courses). Overall survival and early death rate were 94.3% and 0.7%, respectively, for the high-risk GTN cohort, representing a considerable improvement over outcomes reported for an earlier cohort who did not receive induction chemotherapy.<sup>64</sup> Thus, the Panel recommends considering induction low-dose EP (etoposide 100 mg/m<sup>2</sup>/day IV and cisplatin 20 mg/m<sup>2</sup>/day IV) for 1–3 cycles prior to EMA/CO treatment.

### *Management of CNS Metastases*

Additional treatment considerations are recommended for patients with central nervous system (CNS) metastases, who may require emergency intervention to manage intracranial bleeding or elevated intracranial pressure.<sup>5,83</sup> Rates of CNS metastases are low with post-molar GTN, but approximately 20% of patients with choriocarcinoma have CNS involvement.<sup>5,67,83-87</sup> EMA/CO should be modified to include high-dose methotrexate (1000 mg/m<sup>2</sup>) and additional 15-mg oral doses of leucovorin

every 6 hours for 12 doses, starting 32 hours after the start of the methotrexate infusion.<sup>83</sup>

### *Monitoring/Response Assessment During First-Line Therapy*

Monitoring and response assessment during first-line treatment of high-risk GTN is the same as for low-risk GTN. Patients who respond to primary chemotherapy but subsequently experience a low-level hCG plateau, those who have an incomplete response to primary treatment, and those who experience relapse from remission require further treatment (ie, salvage chemotherapy and/or adjuvant surgery).

### *Salvage Chemotherapy*

Approximately 30% to 40% of patients who are high risk will have an incomplete response to first-line therapy or experience relapse from remission.<sup>88,89</sup> Most of these patients have multiple metastases to sites other than the lung and vagina and many will have received inadequate initial therapy.<sup>90,91</sup> Salvage chemotherapy with drug regimens using etoposide and a platinum agent, often combined with surgical resection of persistent tumor, will result in cure of about 80% to 90% of patients with high-risk disease.<sup>92</sup>

EMA/EP or EP/EMA regimens are considered the most appropriate therapy for patients who have responded to EMA/CO but have plateauing low hCG levels or have developed re-elevation of hCG after a complete response to EMA/CO.<sup>93,94</sup> The rate of complete response/remission with EMA/EP for disease resistant to EMA/CO has been reported between 75% and 85%.<sup>75,93-96</sup>

Additional drug combinations containing etoposide, and a platinum agent have been effective in patients who have developed disease resistant to methotrexate-containing regimens. The Panel prefers the use of the



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following four regimens for methotrexate-resistant GTN: TP/TE (paclitaxel and cisplatin alternating weekly with paclitaxel and etoposide), BEP (bleomycin, etoposide, and cisplatin), VIP (etoposide, ifosfamide, and cisplatin), and ICE (ifosfamide, carboplatin, and etoposide).<sup>57,92,95,97,98</sup> Additionally, TIP (paclitaxel, ifosfamide, and cisplatin) has been used as a salvage chemotherapy regimen in germ cell tumors, including those with choriocarcinoma components.<sup>99-102</sup>

Etoposide-platinum-containing regimens require the use of granulocyte colony-stimulating factor (G-CSF) support to prevent neutropenic complications and treatment delays.<sup>92,97,103</sup> The overall success of salvage therapy in this group of patients is about 80%. Factors associated with worse survival outcomes include high hCG at the start of salvage therapy, greater number of metastatic sites, metastases to sites other than the lung and vagina (stage IV), and FIGO score >12.

### *Additional Agents/Regimens with Potential Activity in Treatment-Resistant GTN*

Several additional treatment regimens have been shown to have some activity when treating resistant GTN, including high-dose chemotherapy (HDC) with peripheral hematopoietic cell transplant, immunotherapy, and other chemotherapy regimens. For a subset of patients with resistant disease despite multidrug chemotherapy, HDC with autologous stem cell support has been reported to produce sustained complete responses.<sup>104-108</sup> A retrospective study of 32 patients with refractory choriocarcinoma or poor-prognosis PSTT/ETT who underwent HDC with peripheral blood stem cell support reported a sustained complete response in 7/32 patients, with 13 of 32 patients remaining disease free at the time of analysis following HDC with or without additional therapy.<sup>106</sup>

Pembrolizumab is a monoclonal antibody that inhibits programmed cell death protein 1 (PD-1), which functions as a checkpoint protein for regulation of various immune cells, including T cells with potential

antitumor activity.<sup>109-111</sup> Programmed death ligand 1 (PD-L1) is strongly expressed by GTN.<sup>112,113</sup> Outcomes were reported for four patients with drug-resistant GTN who received pembrolizumab, including two cases of metastatic choriocarcinoma and two cases of metastatic PSTT or mixed PSTT/ETT.<sup>114</sup> All patients had tumors with high levels of PD-L1 expression. Durable response to pembrolizumab was observed in three of the four cases. The patient whose disease did not respond to pembrolizumab had strong PD-L1 tumor expression but an absence of tumor-infiltrating lymphocytes.<sup>114</sup> Based on these data, the NCCN Panel also added nivolumab, another PD-1 inhibitor, to the list of preferred regimens under useful in certain circumstances that may potentially be effective against treatment-resistant GTN. Pembrolizumab and berahyaluronidase alfa-pmph injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.

Avelumab, a PD-L1 inhibitor, may also be effective against treatment-resistant GTN. Results from a phase II study enrolling 15 patients with GTN who experienced disease progression after single-agent chemotherapy suggested that avelumab was effective in normalizing hCG levels in approximately half of the patients.<sup>115</sup>

Gemcitabine, capecitabine, and fluorouracil may also have potential for treating GTN in this setting. Limited data have suggested activity of gemcitabine, administered with or without a platinum agent.<sup>116</sup> Additional support for the potential activity of these regimens in GTN can be found in the data for treating germ cell tumors. Successful use of capecitabine as single-agent salvage chemotherapy has been reported.<sup>117,118</sup> Groups in Asia have also reported on fluorouracil, primarily in combination with dactinomycin.<sup>119</sup>



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Nivolumab and ipilimumab combination is added to the additional agents/regimens shown to have some activity in treating multiagent chemotherapy-resistant GTN based on Patel SP et al.<sup>120</sup> In this prospective, open-label phase II trial that evaluated ipilimumab plus nivolumab across multiple rare tumor cohorts, including GTN, four patients with refractory GTN enrolled and received therapy. Three of four patients achieved ongoing objective responses with a reasonable safety profile at 6 to 11+ months. At 11 months of ongoing follow-up, three of four patients responded (overall response rate [ORR] = 75% [complete response (CR), 25%, n = 1; partial response (PR), 50%, n = 2]). Responders included malignant GTN (n = 1, CR, progression-free survival [PFS] 11+ months) and choriocarcinoma (n = 2, both PRs, PFS 10+ and 6+ months). Ipilimumab plus nivolumab have demonstrated efficacy in chemotherapy-refractory GTN and is recommended as a systemic therapy option to High-Risk GTN under Useful in Certain Circumstances. Following FDA approval<sup>121</sup> of nivolumab and hyaluronidase for subcutaneous injection across approved adult, solid tumor as monotherapy and monotherapy maintenance following completion of nivolumab, the NCCN Panel added that nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.

### **Adjuvant Surgery**

Adjuvant surgical procedures, especially hysterectomy with salpingectomy and pulmonary resection, may be required to manage high-risk, chemotherapy-resistant disease. Select patients with isolated disease, especially in the uterus and lungs, may be candidates for surgical resection.<sup>122-124</sup> PET/CT imaging may be useful for detecting isolated metastatic sites that are amenable to targeted surgery.<sup>125</sup> Additionally, interventional procedures to prevent or control hemorrhage are important components in the management of high-risk GTN.<sup>5</sup> Selective arterial embolization can be used to manage bleeding from the uterus/vagina or other tumor sites.<sup>126-128</sup> In one case series, nearly 50%

of patients with high-risk disease underwent some form of surgical procedure during the course of treatment in order to effect cure.<sup>129</sup>

### **Intermediate Trophoblastic Tumors**

Whereas molar pregnancies and choriocarcinoma are derived from villous trophoblast (ie, cytotrophoblast and syncytiotrophoblast), ITTs (including PSTT and ETT) develop from extravillous trophoblast (ie, intermediate trophoblast). ITTs comprise approximately 1% of GTN cases, and as such, their biologic behavior and treatment are less well established. These tumors typically develop months to years following normal pregnancies but can occur after any pregnancy event. A series of 62 cases of ITT suggested that interval between antecedent pregnancy and disease onset may be longer for ETT than PSTT.<sup>130</sup>

PSTT and ETT are generally slow-growing tumors that can metastasize months or years after initial primary tumor development and often present with abnormal uterine bleeding or amenorrhea. The vast majority of ITTs secrete hCG, but at significantly lower levels compared with other types of GTN. As such, hCG is a less reliable tumor marker for these subtypes of GTN. At diagnosis, metastases are noted in 30% to 50% of cases, most commonly to the lungs. Unlike other GTNs, these have a greater propensity for lymphatic spread. Data are currently being collected in a global database of PSTTs and ETTs through the efforts of the International Society for the Study of Trophoblastic Disease (ISSTD).<sup>11,131-136</sup>

ITTs can be differentiated from other types of GTN via their histopathologic characteristics.<sup>11</sup> In PSTT, IHC staining reveals the diffuse presence of Mel-CAM (CD146) and hPL, whereas hCG staining is only focal. Cytogenetic studies have revealed that PSTTs are more often diploid than aneuploid.<sup>137</sup> Serum hPL measurements are not clinically useful in monitoring disease course or guiding clinical





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management.<sup>134,135,138,139</sup> ETT is distinguished from PSTT by its smaller, fairly monomorphic cells and a nested, nodular, well-circumscribed growth pattern. IHC reveals strong expression of p63, but weak expression of Mel-CAM and hPL.<sup>140</sup> Greater than 50% of cells in ETT also express cyclin E. It frequently involves the lower uterine segment and endocervix, and because of its epithelioid histologic appearance and expression of p63, ETT can be confused with squamous cell carcinoma.<sup>11,140,141</sup>

Due to the rarity of these tumors, generally small cohort sizes preclude rigorous statistical analysis of risk factors in ITT. The FIGO prognostic scoring system for GTN does not correlate well with outcomes in PSTT and ETT.<sup>11</sup> Based on findings from the largest existing database, PSTT and ETT accounted for 125 of 54,743 cases of GTD (0.23%), with post-treatment 5- and 10-year survival estimates of 80% and 75%, respectively. The most important prognostic factors include advanced disease stage and interval from last known pregnancy event of  $\geq 48$  months.<sup>132,135,136,142</sup> Additional risk factors associated with less favorable outcomes are advancing age, deep myometrial invasion, tumor necrosis, large tumor size, and mitotic index.<sup>11,136,143</sup>

### **Treatment Approach**

ITTs are relatively chemoresistant and thus follow a somewhat different treatment paradigm than invasive mole and choriocarcinoma, with surgical intervention playing a more critical role. Treatment of PSTT and ETT is mainly determined by presence or absence of metastatic disease, with some consideration given to high-risk factors. The survival rate is approximately 100% for nonmetastatic disease and 50% to 60% for metastatic disease. Increased use of platinum-based CT over time has led to improved overall survival for the subset of patients with ITT who have an overall poor prognosis (ie, interval  $\geq 48$  months from last known pregnancy event).<sup>132,134-136</sup>

Hysterectomy with salpingectomy, with or without pelvic lymph node biopsy, is the recommended treatment for nonmetastatic (stage I) disease. If the patient has any of the reported poor prognostic factors<sup>137</sup> such as  $\geq 2$  years interval from index pregnancy, deep myometrial invasion, necrosis or increased mitotic count ( $>5/10$  high-power fields), systemic therapy regimens need to be considered. In Version 1.2025, the recommendation of  $\geq 2$  years interval from index pregnancy is changed to  $\geq 4$  years based on a study conducted by Froeling et al.<sup>136</sup> Of 54,743 patients with GTD screened from the United Kingdom GTD database, 125 (0.23%) were diagnosed with PSTT and/or ETT. Probability of survival at 5 and 10 years following treatment was 80% (95% CI, 72.8%–87.6%) and 75% (95% CI, 66.3%–84.3%), respectively. Of the five prognostic factors identified from Univariate analysis for reduced overall survival (age, FIGO stage, time since antecedent pregnancy, hCG level, mitotic index), stage IV disease and interval  $\geq 48$  months since antecedent pregnancy (hazard ratio, 14.57; 95% CI, 4.17–50.96;  $P < .001$ ) were most significant on multivariable analysis. PSTT/ETT with advanced FIGO stage or an interval  $\geq 48$  months from their last known pregnancy have been shown to have poorer outcomes and the Panel changed the prognostic factor to  $\geq 4$ -year interval from causative gestation. Hysterectomy with salpingectomy and chemotherapy are recommended for metastatic disease. Metastasectomy (if feasible) should also be performed for isolated distant disease, especially in the lungs. EMA/EP and EP/EMA and TP/TE are the preferred chemotherapy regimens for patients with metastatic disease; other recommended regimens that are effective against treatment-resistant GTN such as BEP, VIP, and ICE may also have some efficacy in metastatic ITTs. All of these regimens may also be considered for patients with nonmetastatic disease who have one or more adverse prognostic factors (ie, interval from causative gestation  $\geq 4$  years).<sup>144,145</sup> Additional agents/regimens shown to have some activity in treating multiagent chemotherapy-resistant GTN including PD-1/PD-L1 inhibitors, capecitabine- based regimens, gemcitabine with or without cisplatin, high-



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dose chemotherapy with peripheral HCT and nivolumab and ipilimumab combinations are also recommended as useful in other circumstances for ITTs.

### *Monitoring and Surveillance*

Post-treatment hCG levels should be monitored as indicated for GTN, although hCG is a less reliable tumor marker for ITTs versus post-molar GTN. The Panel recommends surveillance with FDG-PET/CT at the completion of chemotherapy and then every 6 to 12 months for 2 to 3 years.

### *Post-Treatment Recurrence or Progression of ITTs*

Systemic therapy regimens recommended for primary treatment of metastatic disease may have some efficacy in patients experiencing recurrence or progression of nonmetastatic or metastatic ITTs. Alternatively, patients with metastatic ITTs may forego additional chemotherapy and instead receive best supportive care (see NCCN Guidelines for Palliative Care, available at [www.NCCN.org](http://www.NCCN.org)).

## Gynecologic Survivorship

Treatment for gynecologic cancer typically involves surgery, chemotherapy, hormone therapy, radiation therapy, and/or immunotherapy, which may cause acute, short- and long-term toxicities. Surgical approaches may be extensive and cause adhesions to form, which in turn may cause pain and contribute to the development of small bowel obstruction, urinary or gastrointestinal complications (eg, incontinence, diarrhea), pelvic floor dysfunction (manifested by a variety of urinary, bowel, and/or sexual effects), and lymphedema.<sup>146,147</sup> Chemotherapy agents vary, though commonly used regimens may pose a significant risk of neurotoxicity, cardiac toxicity, cognitive dysfunction, and the development of hematologic cancers.<sup>148</sup> Long-term estrogen deprivation may cause symptoms such as hot flashes, vaginal dryness,

and bone loss. Radiation therapy may cause long-term complications (eg, fibrosis, stenosis, vulvovaginal atrophy)<sup>149,150</sup> and may predispose patients to subsequent cancers of the skin, subcutaneous tissue, and/or underlying organs that are proximal to the radiation field.<sup>151</sup> Use of immunotherapy agents in gynecologic cancers is emerging, and to date, long-term effects of these treatments are unknown.<sup>152,153</sup>

Following completion of treatment, all gynecologic cancer survivors should receive regular general medical care that focuses on managing chronic diseases (eg, depression, diabetes, hypertension), monitoring cardiovascular risk factors, receiving recommended vaccinations, and encouraging adoption of a healthy lifestyle (eg, promoting exercise, smoking cessation).<sup>154,155</sup> In order to assess the late and long-term effects of gynecologic cancers, clinicians should comprehensively document the patient's history, including prior treatment history, and conduct a thorough physical examination followed by necessary imaging and/or laboratory testing.<sup>155</sup> As most treatments for gynecologic cancers will cause sexual dysfunction, early menopause, and infertility, special attention to the resultant medical and psychosocial implications is needed. All individuals, whether sexually active or not, should be asked about genitourinary symptoms, including vulvovaginal dryness.<sup>156</sup> Post-radiation use of vaginal dilators and moisturizers is recommended.<sup>149,157</sup> Psychosocial effects may include psychological (eg, depression, anxiety, fear of recurrence, altered body image), financial (eg, return to work, insurance concerns), and interpersonal (eg, relationships, sexuality, intimacy).<sup>155</sup> Patients should be referred to appropriate specialty providers (eg, physical therapy, pelvic floor therapy, sexual therapy, psychotherapy) as needed, based on prior treatment history and assessed risk of developing late effects and/or existing concerns.

Communication and coordination with all clinicians involved in the care of survivors, including primary care clinicians, is critical.<sup>155,158</sup> During



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subsequent pregnancies, early ultrasound to confirm intrauterine pregnancy is recommended. Pathologic evaluation of placenta needs to be performed and hCG levels need to be checked at 6 weeks postpartum to confirm that the levels have returned to negative. Providing survivors with a summary of their treatment and recommendations for follow-up is also recommended. To this end, the Society of Gynecologic Oncology (SGO) has developed templates for gynecologic cancer-specific Survivorship Care Plans to aid survivors and their clinicians in summarizing cancer history, treatments received, possible side effects, and recommended follow-up.<sup>159</sup>



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### References

1. Altieri A, Franceschi S, Ferlay J, et al. Epidemiology and aetiology of gestational trophoblastic diseases. *Lancet Oncol* 2003;4:670–678. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/14602247>.
2. Lurain JR. Gestational trophoblastic disease I: epidemiology, pathology, clinical presentation and diagnosis of gestational trophoblastic disease, and management of hydatidiform mole. *Am J Obstet Gynecol* 2010;203:531–539. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20728069>.
3. Seckl MJ, Sebire NJ, Berkowitz RS. Gestational trophoblastic disease. *Lancet* 2010;376:717–729. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20673583>.
4. Chawla T, Bouchard-Fortier G, Turashvili G, et al. Gestational trophoblastic disease: an update. *Abdom Radiol (NY)* 2023;48:1793–1815. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/36763119>.
5. Brown J, Naumann RW, Seckl MJ, Schink J. 15years of progress in gestational trophoblastic disease: Scoring, standardization, and salvage. *Gynecol Oncol* 2017;144:200–207. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27743739>.
6. Lurain JR. Gestational trophoblastic disease II: classification and management of gestational trophoblastic neoplasia. *Am J Obstet Gynecol* 2011;204:11–18. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20739008>.
7. Berkowitz RS, Goldstein DP. Current advances in the management of gestational trophoblastic disease. *Gynecol Oncol* 2013;128:3–5. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/22846466>.
8. Berkowitz RS, Goldstein DP. Chorionic tumors. *N Engl J Med* 1996;335:1740–1748. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/8929267>.
9. Berkowitz RS, Goldstein DP. Clinical practice. Molar pregnancy. *N Engl J Med* 2009;360:1639–1645. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19369669>.
10. Sun SY, Melamed A, Goldstein DP, et al. Changing presentation of complete hydatidiform mole at the New England Trophoblastic Disease Center over the past three decades: does early diagnosis alter risk for gestational trophoblastic neoplasia? *Gynecol Oncol* 2015;138:46–49. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/25969351>.
11. Horowitz NS, Goldstein DP, Berkowitz RS. Placental site trophoblastic tumors and epithelioid trophoblastic tumors: Biology, natural history, and treatment modalities. *Gynecol Oncol* 2017;144:208–214. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27789086>.
12. Shaaban AM, Rezvani M, Haroun RR, et al. Gestational Trophoblastic Disease: Clinical and Imaging Features. *Radiographics* 2017;37:681–700. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/28287945>.
13. Hancock BW, Tidy JA. Current management of molar pregnancy. *J Reprod Med* 2002;47:347–354. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/12063873>.
14. Ngan HY, Seckl MJ, Berkowitz RS, et al. Update on the diagnosis and management of gestational trophoblastic disease. *Int J Gynaecol Obstet* 2015;131 Suppl 2:S123–126. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26433668>.
15. Zhao P, Lu Y, Huang W, et al. Total hysterectomy versus uterine evacuation for preventing post-molar gestational trophoblastic neoplasia in patients who are at least 40 years old: a systematic review and meta-analysis. *BMC Cancer* 2019;19:13. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30612545>.
16. Kurman RJ, Carcangiu ML, Herrington CS, Young RH. WHO Classification of Tumours of Female Reproductive Organs. Fourth Edition. Vol. 6: World Health Organization; 2014.





## NCCN Guidelines Version 2.2026

### Gestational Trophoblastic Neoplasia

17. Sita-Lumsden A, Short D, Lindsay I, et al. Treatment outcomes for 618 women with gestational trophoblastic tumours following a molar pregnancy at the Charing Cross Hospital, 2000-2009. *Br J Cancer* 2012;107:1810–1814. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23059744>.
18. Wang Q, Fu J, Hu L, et al. Prophylactic chemotherapy for hydatidiform mole to prevent gestational trophoblastic neoplasia. *Cochrane Database Syst Rev* 2017;9:CD007289. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/28892119>.
19. Berkowitz RS, Goldstein DP. Gestational trophoblastic disease. *Cancer* 1995;76:2079–2085. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/8635004>.
20. Cole LA. Human chorionic gonadotropin and associated molecules. *Expert Rev Mol Diagn* 2009;9:51–73. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19099349>.
21. Ngan HYS, Seckl MJ, Berkowitz RS, et al. Update on the diagnosis and management of gestational trophoblastic disease. *Int J Gynaecol Obstet* 2018;143 Suppl 2:79–85. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30306586>.
22. Muller CY, Cole LA. The quagmire of hCG and hCG testing in gynecologic oncology. *Gynecol Oncol* 2009;112:663–672. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19007977>.
23. Coyle C, Short D, Jackson L, et al. What is the optimal duration of human chorionic gonadotrophin surveillance following evacuation of a molar pregnancy? A retrospective analysis on over 20,000 consecutive patients. *Gynecol Oncol* 2018;148:254–257. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/29229282>.
24. Schmitt C, Doret M, Massardier J, et al. Risk of gestational trophoblastic neoplasia after hCG normalisation according to hydatidiform mole type. *Gynecol Oncol* 2013;130:86–89. Available at:
25. Braga A, Maesta I, Matos M, et al. Gestational trophoblastic neoplasia after spontaneous human chorionic gonadotropin normalization following molar pregnancy evacuation. *Gynecol Oncol* 2015;139:283–287. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26383828>.
26. Swift BE, Coopmans L, Singh K, et al. Monitoring complete hydatidiform molar pregnancies after normalisation of human chorionic gonadotrophin: national retrospective population study. *BMJ Med* 2025;4:e001017. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/40309224>.
27. Lavie I, Rao GG, Castrillon DH, et al. Duration of human chorionic gonadotropin surveillance for partial hydatidiform moles. *Am J Obstet Gynecol* 2005;192:1362–1364. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/15902109>.
28. Wolfberg AJ, Growdon WB, Feltmate CM, et al. Low risk of relapse after achieving undetectable HCG levels in women with partial molar pregnancy. *Obstet Gynecol* 2006;108:393–396. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16880311>.
29. Horowitz NS, Eskander RN, Adelman MR, Burke W. Epidemiology, diagnosis, and treatment of gestational trophoblastic disease: A Society of Gynecologic Oncology evidenced-based review and recommendation. *Gynecologic Oncology* 2021;163:605–613. Available at: <https://doi.org/10.1016/j.ygyno.2021.10.003>.
30. Committee FO. FIGO staging for gestational trophoblastic neoplasia 2000. FIGO Oncology Committee. *Int J Gynaecol Obstet* 2002;77:285–287. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/12065144>.
31. Amin MB, Greene FL, Edge SB, et al. The Eighth Edition AJCC Cancer Staging Manual: Continuing to build a bridge from a population-based to a more "personalized" approach to cancer staging. *CA Cancer J Clin* 2017;67:93–99. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/28094848>.



## NCCN Guidelines Version 2.2026

### Gestational Trophoblastic Neoplasia

32. Eysbouts YK, Massuger L, Int'Hout J, et al. The added value of hysterectomy in the management of gestational trophoblastic neoplasia. *Gynecol Oncol* 2017;145:536–542. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/28390821>.

33. Doll KM, Soper JT. The role of surgery in the management of gestational trophoblastic neoplasia. *Obstet Gynecol Surv* 2013;68:533–542. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23803756>.

34. Pezeshki M, Hancock BW, Silcocks P, et al. The role of repeat uterine evacuation in the management of persistent gestational trophoblastic disease. *Gynecol Oncol* 2004;95:423–429. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/15581942>.

35. Garner EI, Feltmate CM, Goldstein DP, Berkowitz RS. The curative effect of a second curettage in persistent trophoblastic disease: a retrospective cohort survey. *Gynecol Oncol* 2005;99:3–5. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16139344>.

36. van Trommel NE, Thomas CM, Massuger LF, Sweep FC. Second curettage in persistent trophoblastic disease (PTD): the need for univocal definition of PTD. *Gynecol Oncol* 2005;99:250–251; author reply 251. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16224821>.

37. Osborne RJ, Filiaci VL, Schink JC, et al. Second Curettage for Low-Risk Nonmetastatic Gestational Trophoblastic Neoplasia. *Obstet Gynecol* 2016;128:535–542. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27500329>.

38. Savage P, Seckl MJ. The role of repeat uterine evacuation in trophoblast disease. *Gynecol Oncol* 2005;99:251–252; author reply 252–253. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16137746>.

39. May T, Goldstein DP, Berkowitz RS. Current chemotherapeutic management of patients with gestational trophoblastic neoplasia. *Chemother Res Pract* 2011;2011:806256. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/22312558>.

40. Rotmensch S, Cole LA. False diagnosis and needless therapy of presumed malignant disease in women with false-positive human chorionic gonadotropin concentrations. *Lancet* 2000;355:712–715. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/10703803>.

41. Cole LA, Muller CY. Hyperglycosylated hCG in the management of quiescent and chemorefractory gestational trophoblastic diseases. *Gynecol Oncol* 2010;116:3–9. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19822356>.

42. Ngan HY, Bender H, Benedet JL, et al. Gestational trophoblastic neoplasia, FIGO 2000 staging and classification. *Int J Gynaecol Obstet* 2003;83 Suppl 1:175–177. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/14763174>.

43. Gestational trophoblastic diseases. Report of a WHO Scientific Group. *World Health Organ Tech Rep Ser* 1983;692:7–81. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/6318459>.

44. Bagshawe KD. Risk and prognostic factors in trophoblastic neoplasia. *Cancer* 1976;38:1373–1385. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/182354>.

45. Goldstein DP, Berkowitz RS, Horowitz NS. Optimal management of low-risk gestational trophoblastic neoplasia. *Expert Rev Anticancer Ther* 2015;15:1293–1304. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26517533>.

46. Chi D, Berchuck A, Dizon D, Yashar C. Table 26.2: Immunohistochemical Markers for Differential Diagnosis of GTD. *Principles and Practice of Gynecologic Oncology* 2017:745.

47. Krishnamurti U, Movahedi-Lankarani S, Bell D, et al. College of American Pathologists Protocol for the Examination of Specimens From Patients With Primary Gestational Trophoblastic Malignancy. Vol. 4.0.0.0; 2017. Available at: <https://documents.cap.org/protocols/cp-trophoblast-17protocol-4000.pdf>.





## NCCN Guidelines Version 2.2026

### Gestational Trophoblastic Neoplasia

48. Bolze PA, Mathe M, Hajri T, et al. First-line hysterectomy for women with low-risk non-metastatic gestational trophoblastic neoplasia no longer wishing to conceive. *Gynecol Oncol* 2018;150:282–287. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/29887485>.
49. Mangili G, Lorusso D, Brown J, et al. Trophoblastic disease review for diagnosis and management: a joint report from the International Society for the Study of Trophoblastic Disease, European Organisation for the Treatment of Trophoblastic Disease, and the Gynecologic Cancer InterGroup. *Int J Gynecol Cancer* 2014;24:S109–116. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/25341573>.
50. Lawrie TA, Alazzam M, Tidy J, et al. First-line chemotherapy in low-risk gestational trophoblastic neoplasia. *Cochrane Database Syst Rev* 2016;2016:CD007102. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27281496>.
51. Schink JC, Filiaci V, Huang HQ, et al. An international randomized phase III trial of pulse actinomycin-D versus multi-day methotrexate for the treatment of low risk gestational trophoblastic neoplasia; NRG/GOG 275. *Gynecol Oncol* 2020;158:354–360. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/32460997>.
52. Osborne RJ, Filiaci V, Schink JC, et al. Phase III trial of weekly methotrexate or pulsed dactinomycin for low-risk gestational trophoblastic neoplasia: a gynecologic oncology group study. *J Clin Oncol* 2011;29:825–831. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/21263100>.
53. Kohorn EI. Is lack of response to single-agent chemotherapy in gestational trophoblastic disease associated with dose scheduling or chemotherapy resistance? *Gynecol Oncol* 2002;85:36–39. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/11925117>.
54. Hitchins RN, Holden L, Newlands ES, et al. Single agent etoposide in gestational trophoblastic tumours. Experience at Charing Cross Hospital 1978-1987. *Eur J Cancer Clin Oncol* 1988;24:1041–1046. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/2842160>.
55. Sung HC, Wu PC, Yang HY. Reevaluation of 5-fluorouracil as a single therapeutic agent for gestational trophoblastic neoplasms. *Am J Obstet Gynecol* 1984;150:69–75. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/6089563>.
56. Lybol C, Sweep FC, Harvey R, et al. Relapse rates after two versus three consolidation courses of methotrexate in the treatment of low-risk gestational trophoblastic neoplasia. *Gynecol Oncol* 2012;125:576–579. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/22410329>.
57. Alazzam M, Tidy J, Osborne R, et al. Chemotherapy for resistant or recurrent gestational trophoblastic neoplasia. *Cochrane Database Syst Rev* 2016;2016:CD008891. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26760424>.
58. Cortés-Charry R, Hennah L, Froeling FEM, et al. Increasing the human chorionic gonadotrophin cut-off to  $\leq 1000$  IU/l for starting actinomycin D in post-molar gestational trophoblastic neoplasia developing resistance to methotrexate spares more women multi-agent chemotherapy. *ESMO Open* 2021;6:100110. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/33845362>.
59. Covens A, Filiaci VL, Burger RA, et al. Phase II trial of pulse dactinomycin as salvage therapy for failed low-risk gestational trophoblastic neoplasia: a Gynecologic Oncology Group study. *Cancer* 2006;107:1280–1286. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16900525>.
60. McNeish IA, Strickland S, Holden L, et al. Low-risk persistent gestational trophoblastic disease: outcome after initial treatment with low-dose methotrexate and folinic acid from 1992 to 2000. *J Clin Oncol* 2002;20:1838–1844. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/11919242>.
61. McGrath S, Short D, Harvey R, et al. The management and outcome of women with post-hydatidiform mole 'low-risk' gestational trophoblastic neoplasia, but hCG levels in excess of 100 000 IU l(-1). *Br J Cancer* 2010;102:810–814. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20160727>.



## NCCN Guidelines Version 2.2026

### Gestational Trophoblastic Neoplasia

62. Lurain JR, Chapman-Davis E, Hoekstra AV, Schink JC. Actinomycin D for methotrexate-failed low-risk gestational trophoblastic neoplasia. *J Reprod Med* 2012;57:283–287. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/22838241>.

63. Prouvot C, Golfier F, Massardier J, et al. Efficacy and Safety of Second-Line 5-Day Dactinomycin in Case of Methotrexate Failure for Gestational Trophoblastic Neoplasia. *Int J Gynecol Cancer* 2018;28:1038–1044. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/29629964>.

64. Alifrangis C, Agarwal R, Short D, et al. EMA/CO for high-risk gestational trophoblastic neoplasia: good outcomes with induction low-dose etoposide-cisplatin and genetic analysis. *J Clin Oncol* 2013;31:280–286. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23233709>.

65. Lurain JR, Singh DK, Schink JC. Primary treatment of metastatic high-risk gestational trophoblastic neoplasia with EMA-CO chemotherapy. *J Reprod Med* 2006;51:767–772. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17086804>.

66. Ahamed E, Short D, North B, et al. Survival of women with gestational trophoblastic neoplasia and liver metastases: is it improving? *J Reprod Med* 2012;57:262–269. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/22696824>.

67. Newlands ES, Holden L, Seckl MJ, et al. Management of brain metastases in patients with high-risk gestational trophoblastic tumors. *J Reprod Med* 2002;47:465–471. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/12092015>.

68. Crawford RA, Newlands E, Rustin GJ, et al. Gestational trophoblastic disease with liver metastases: the Charing Cross experience. *Br J Obstet Gynaecol* 1997;104:105–109. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/8988707>.

69. Newlands ES, Bagshawe KD, Begent RH, et al. Results with the EMA/CO (etoposide, methotrexate, actinomycin D, cyclophosphamide,

vincristine) regimen in high risk gestational trophoblastic tumours, 1979 to 1989. *Br J Obstet Gynaecol* 1991;98:550–557. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/1651757>.

70. Bolis G, Bonazzi C, Landoni F, et al. EMA/CO regimen in high-risk gestational trophoblastic tumor (GTT). *Gynecol Oncol* 1988;31:439–444. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/2846414>.

71. Kim SJ, Bae SN, Kim JH, et al. Risk factors for the prediction of treatment failure in gestational trophoblastic tumors treated with EMA/CO regimen. *Gynecol Oncol* 1998;71:247–253. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9826467>.

72. Matsui H, Suzuka K, Iitsuka Y, et al. Combination chemotherapy with methotrexate, etoposide, and actinomycin D for high-risk gestational trophoblastic tumors. *Gynecol Oncol* 2000;78:28–31. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/10873405>.

73. Escobar PF, Lurain JR, Singh DK, et al. Treatment of high-risk gestational trophoblastic neoplasia with etoposide, methotrexate, actinomycin D, cyclophosphamide, and vincristine chemotherapy. *Gynecol Oncol* 2003;91:552–557. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/14675675>.

74. Turan T, Karacay O, Tulunay G, et al. Results with EMA/CO (etoposide, methotrexate, actinomycin D, cyclophosphamide, vincristine) chemotherapy in gestational trophoblastic neoplasia. *Int J Gynecol Cancer* 2006;16:1432–1438. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16803542>.

75. Lu WG, Ye F, Shen YM, et al. EMA-CO chemotherapy for high-risk gestational trophoblastic neoplasia: a clinical analysis of 54 patients. *Int J Gynecol Cancer* 2008;18:357–362. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17711444>.

76. Cagayan MS. High-risk metastatic gestational trophoblastic neoplasia. Primary management with EMA-CO (etoposide, methotrexate, actinomycin D, cyclophosphamide and vincristine) chemotherapy. *J*



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

Reprod Med 2012;57:231–236. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/22696818>.

77. Cyriac S, Rajendranath R, Sridevi V, Sagar TG. Etoposide, cisplatin-etoposide, methotrexate, actinomycin-D as primary treatment for management of very-high-risk gestational trophoblastic neoplasia. Int J Gynaecol Obstet 2011;115:37–39. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/21802685>.

78. Ghaemmaghami F, Modares M, Arab M, et al. EMA-EP regimen, as firstline multiple agent chemotherapy in high-risk GTT patients (stage II–IV). Int J Gynecol Cancer 2004;14:360–365. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/15086738>.

79. Han SN, Amant F, Leunen K, et al. EP-EMA regimen (etoposide and cisplatin with etoposide, methotrexate, and dactinomycin) in a series of 18 women with gestational trophoblastic neoplasia. Int J Gynecol Cancer 2012;22:875–880. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/22635033>.

80. Deng L, Yan X, Zhang J, Wu T. Combination chemotherapy for high-risk gestational trophoblastic tumour. Cochrane Database Syst Rev 2009;CD005196. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/19370618>.

81. Kong Y, Yang J, Jiang F, et al. Clinical characteristics and prognosis of ultra high-risk gestational trophoblastic neoplasia patients: A retrospective cohort study. Gynecol Oncol 2017;146:81–86. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/28461032>.

82. Bolze PA, Riedl C, Massardier J, et al. Mortality rate of gestational trophoblastic neoplasia with a FIGO score of  $\geq 13$ . Am J Obstet Gynecol 2016;214:390.e1–8. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/26433171>.

83. Savage P, Kelpanides I, Tuthill M, et al. Brain metastases in gestational trophoblast neoplasia: an update on incidence, management and outcome. Gynecol Oncol 2015;137:73–76. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/25598530>.

84. Gavanier D, Leport H, Massardier J, et al. Gestational trophoblastic neoplasia with brain metastasis at initial presentation: a retrospective study. Int J Clin Oncol 2019;24:153–160. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/30242539>.

85. Piura E, Piura B. Brain metastases from gestational trophoblastic neoplasia: review of pertinent literature. Eur J Gynaecol Oncol 2014;35:359–367. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/25118474>.

86. Neubauer NL, Latif N, Kalakota K, et al. Brain metastasis in gestational trophoblastic neoplasia: an update. J Reprod Med 2012;57:288–292. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/22838242>.

87. Soper JT, Spillman M, Sampson JH, et al. High-risk gestational trophoblastic neoplasia with brain metastases: individualized multidisciplinary therapy in the management of four patients. Gynecol Oncol 2007;104:691–694. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/17137617>.

88. Powles T, Savage PM, Stebbing J, et al. A comparison of patients with relapsed and chemo-refractory gestational trophoblastic neoplasia. Br J Cancer 2007;96:732–737. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/17299394>.

89. Hoekstra AV, Lurain JR, Rademaker AW, Schink JC. Gestational trophoblastic neoplasia: treatment outcomes. Obstet Gynecol 2008;112:251–258. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/18669719>.

90. Lurain JR, Casanova LA, Miller DS, Rademaker AW. Prognostic factors in gestational trophoblastic tumors: a proposed new scoring system based on multivariate analysis. Am J Obstet Gynecol 1991;164:611–616. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/1847005>.

91. Lurain JR, Hoekstra AV, Schink JC. Results of treatment of patients with gestational trophoblastic neoplasia referred to the Brewer



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

Trophoblastic Disease Center after failure of treatment elsewhere (1979-2006). J Reprod Med 2008;53:535–540. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/18720930>.

92. Lurain JR, Schink JC. Importance of salvage therapy in the management of high-risk gestational trophoblastic neoplasia. J Reprod Med 2012;57:219–224. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/22696816>.

93. Newlands ES, Mulholland PJ, Holden L, et al. Etoposide and cisplatin/etoposide, methotrexate, and actinomycin D (EMA) chemotherapy for patients with high-risk gestational trophoblastic tumors refractory to EMA/cyclophosphamide and vincristine chemotherapy and patients presenting with metastatic placental site trophoblastic tumors. J Clin Oncol 2000;18:854–859. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/10673528>.

94. Mao Y, Wan X, Lv W, Xie X. Relapsed or refractory gestational trophoblastic neoplasia treated with the etoposide and cisplatin/etoposide, methotrexate, and actinomycin D (EP-EMA) regimen. Int J Gynaecol Obstet 2007;98:44–47. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/17481633>.

95. Anantharaju AA, Pallavi VR, Bafna UD, et al. Role of salvage therapy in chemo resistant or recurrent high-risk gestational trophoblastic neoplasm. Int J Gynecol Cancer 2019;29:547–553. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/30700567>.

96. Lurain JR, Singh DK, Schink JC. Management of metastatic high-risk gestational trophoblastic neoplasia: FIGO stages II-IV: risk factor score > or = 7. J Reprod Med 2010;55:199–207. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/20626175>.

97. Wang J, Short D, Sebire NJ, et al. Salvage chemotherapy of relapsed or high-risk gestational trophoblastic neoplasia (GTN) with paclitaxel/cisplatin alternating with paclitaxel/etoposide (TP/TE). Ann Oncol 2008;19:1578–1583. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/18453518>.

98. Essel KG, Bruegl A, Gershenson DM, et al. Salvage chemotherapy for gestational trophoblastic neoplasia: Utility or futility? Gynecol Oncol 2017;146:74–80. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/28473205>.

99. Feldman DR, Hu J, Dorff TB, et al. Paclitaxel, Ifosfamide, and Cisplatin Efficacy for First-Line Treatment of Patients With Intermediate- or Poor-Risk Germ Cell Tumors. J Clin Oncol 2016;34:2478–2483. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/27185842>.

100. Mardiak J, Rejlekova K, Mego M, et al. Determination of efficacy of TIP combination (paclitaxel, ifosfamide, cisplatin) as first salvage therapy for patients with relapsed germ cell tumors in a poor prognosis group [abstract]. J Clin Oncol 2009;27:Abstract e16049. Available at:

[http://ascopubs.org/doi/abs/10.1200/jco.2009.27.15\\_suppl.e16049](http://ascopubs.org/doi/abs/10.1200/jco.2009.27.15_suppl.e16049).

101. Kondagunta GV, Bacik J, Donadio A, et al. Combination of paclitaxel, ifosfamide, and cisplatin is an effective second-line therapy for patients with relapsed testicular germ cell tumors. J Clin Oncol 2005;23:6549–6555. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/16170162>.

102. Motzer RJ, Sheinfeld J, Mazumdar M, et al. Paclitaxel, ifosfamide, and cisplatin second-line therapy for patients with relapsed testicular germ cell cancer. J Clin Oncol 2000;18:2413–2418. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/10856101>.

103. Lurain JR, Nejad B. Secondary chemotherapy for high-risk gestational trophoblastic neoplasia. Gynecol Oncol 2005;97:618–623. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/15863169>.

104. Benigno BB. High-dose chemotherapy with autologous stem cell support as salvage therapy in recurrent gestational trophoblastic disease. Int J Gynecol Cancer 2013;23:1331–1333. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/23970157>.

105. El-Helw LM, Seckl MJ, Haynes R, et al. High-dose chemotherapy and peripheral blood stem cell support in refractory gestational





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trophoblastic neoplasia. Br J Cancer 2005;93:620–621. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16222307>.

106. Frijstein MM, Lok CAR, Short D, et al. The results of treatment with high-dose chemotherapy and peripheral blood stem cell support for gestational trophoblastic neoplasia. Eur J Cancer 2019;109:162–171. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30731277>.

107. van Besien K, Verschraegen C, Mehra R, et al. Complete remission of refractory gestational trophoblastic disease with brain metastases treated with multicycle ifosfamide, carboplatin, and etoposide (ICE) and stem cell rescue. Gynecol Oncol 1997;65:366–369. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9159354>.

108. Yamamoto E, Niimi K, Fujikake K, et al. High-dose chemotherapy with autologous peripheral blood stem cell transplantation for choriocarcinoma: A case report and literature review. Mol Clin Oncol 2016;5:660–664. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27900108>.

109. Buchbinder EI, Desai A. CTLA-4 and PD-1 Pathways: Similarities, Differences, and Implications of Their Inhibition. Am J Clin Oncol 2016;39:98–106. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26558876>.

110. Chen DS, Irving BA, Hodi FS. Molecular pathways: next-generation immunotherapy--inhibiting programmed death-ligand 1 and programmed death-1. Clin Cancer Res 2012;18:6580–6587. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23087408>.

111. Wherry EJ. T cell exhaustion. Nat Immunol 2011;12:492–499. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/21739672>.

112. Bolze PA, Patrier S, Massardier J, et al. PD-L1 Expression in Premalignant and Malignant Trophoblasts From Gestational Trophoblastic Diseases Is Ubiquitous and Independent of Clinical Outcomes. Int J Gynecol Cancer 2017;27:554–561. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/28060141>.

113. Veras E, Kurman RJ, Wang TL, Shih IM. PD-L1 Expression in Human Placentas and Gestational Trophoblastic Diseases. Int J Gynecol Pathol 2017;36:146–153. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27362903>.

114. Ghorani E, Kaur B, Fisher RA, et al. Pembrolizumab is effective for drug-resistant gestational trophoblastic neoplasia. Lancet 2017;390:2343–2345. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/29185430>.

115. You B, Bolze PA, Lotz JP, et al. Avelumab in Patients With Gestational Trophoblastic Tumors With Resistance to Single-Agent Chemotherapy: Cohort A of the TROPHIMMUN Phase II Trial. J Clin Oncol 2020;38:3129–3137. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/32716740>.

116. Pandian Z, Seckl MJ, Smith R, Lees DA. Gestational choriocarcinoma: an unusual presentation with response to gemcitabine and surgery. BJOG 2004;111:382–384. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/15008778>.

117. Bianconi MI, Otero S, Storino C, Jankilevich G. Role of Capecitabine in the Management of Gestational Trophoblastic Neoplasia: A Drug for Two Settings. J Reprod Med 2017;62:250–256. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30027717>.

118. Bianconi M, Jankilevich G, Otero S, et al. Successful salvage of a relapsed high risk gestational trophoblastic neoplasia patient using capecitabine. Gynecol Oncol 2007;106:268–271. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17493670>.

119. Zhao Y, Zhang W, Duan W. Management of gestational trophoblastic neoplasia with 5-fluorouracil and actinomycin D in northern China. J Reprod Med 2009;54:88–94. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19301571>.

120. Patel SP, Othus M, Chae YK, et al. A Phase II Basket Trial of Dual Anti-CTLA-4 and Anti-PD-1 Blockade in Rare Tumors (DART SWOG 1609 Cohort 47) in Patients with Gestational Trophoblastic Neoplasia.



# NCCN Guidelines Version 2.2026

## Gestational Trophoblastic Neoplasia

Clin Cancer Res 2024;30:33–38. Available at:  
<https://doi.org/10.1158/1078-0432.CCR-23-2293>.

121. FDA approves nivolumab and hyaluronidase-nvhy for subcutaneous injection. 2024. Available at: <https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-nivolumab-and-hyaluronidase-nvhy-subcutaneous-injection>. Accessed 1/31/2025.

122. Alazzam M, Hancock BW, Tidy J. Role of hysterectomy in managing persistent gestational trophoblastic disease. J Reprod Med 2008;53:519–524. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/18720927>.

123. Fleming EL, Garrett L, Growdon WB, et al. The changing role of thoracotomy in gestational trophoblastic neoplasia at the New England Trophoblastic Disease Center. J Reprod Med 2008;53:493–498. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/18720924>.

124. Kanis MJ, Lurain JR. Pulmonary Resection in the Management of High-Risk Gestational Trophoblastic Neoplasia. Int J Gynecol Cancer 2016;26:796–800. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26905332>.

125. Mapelli P, Mangili G, Picchio M, et al. Role of 18F-FDG PET in the management of gestational trophoblastic neoplasia. Eur J Nucl Med Mol Imaging 2013;40:505–513. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23314259>.

126. Lim AK, Agarwal R, Seckl MJ, et al. Embolization of bleeding residual uterine vascular malformations in patients with treated gestational trophoblastic tumors. Radiology 2002;222:640–644. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/11867779>.

127. Tse KY, Chan KK, Tam KF, Ngan HY. 20-year experience of managing profuse bleeding in gestational trophoblastic disease. J Reprod Med 2007;52:397–401. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17583238>.

128. McGrath S, Harding V, Lim AK, et al. Embolization of uterine arteriovenous malformations in patients with gestational trophoblastic

tumors: a review of patients at Charing Cross Hospital, 2000–2009. J Reprod Med 2012;57:319–324. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/22838248>.

129. Lurain JR, Singh DK, Schink JC. Role of surgery in the management of high-risk gestational trophoblastic neoplasia. J Reprod Med 2006;51:773–776. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17086805>.

130. Zhang Y, Zhang S, Huang W, et al. Intermediate trophoblastic tumor: the clinical analysis of 62 cases and prognostic factors. Arch Gynecol Obstet 2019;299:1353–1364. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30607597>.

131. Lan C, Li Y, He J, Liu J. Placental site trophoblastic tumor: lymphatic spread and possible target markers. Gynecol Oncol 2010;116:430–437. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19910023>.

132. Frijstein MM, Lok CAR, van Trommel NE, et al. Management and prognostic factors of epithelioid trophoblastic tumors: Results from the International Society for the Study of Trophoblastic Diseases database. Gynecol Oncol 2019;152:361–367. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30473257>.

133. Kurman RJ, Shih le M. Discovery of a cell: reflections on the checkered history of intermediate trophoblast and update on its nature and pathologic manifestations. Int J Gynecol Pathol 2014;33:339–347. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/24901393>.

134. Schmid P, Nagai Y, Agarwal R, et al. Prognostic markers and long-term outcome of placental-site trophoblastic tumours: a retrospective observational study. Lancet 2009;374:48–55. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/19552948>.

135. Zhao J, Lv WG, Feng FZ, et al. Placental site trophoblastic tumor: A review of 108 cases and their implications for prognosis and treatment. Gynecol Oncol 2016;142:102–108. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27168005>.





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136. Froeling FEM, Ramaswami R, Papanastasopoulos P, et al. Intensified therapies improve survival and identification of novel prognostic factors for placental-site and epithelioid trophoblastic tumours. *Br J Cancer* 2019;120:587–594. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30792530>.

137. Baergen RN, Rutgers JL, Young RH, et al. Placental site trophoblastic tumor: A study of 55 cases and review of the literature emphasizing factors of prognostic significance. *Gynecol Oncol* 2006;100:511–520. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16246400>.

138. Papadopoulos AJ, Foskett M, Seckl MJ, et al. Twenty-five years' clinical experience with placental site trophoblastic tumors. *J Reprod Med* 2002;47:460–464. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/12092014>.

139. Hassadia A, Gillespie A, Tidy J, et al. Placental site trophoblastic tumour: clinical features and management. *Gynecol Oncol* 2005;99:603–607. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16085293>.

140. Shih IM, Kurman RJ. Epithelioid trophoblastic tumor: a neoplasm distinct from choriocarcinoma and placental site trophoblastic tumor simulating carcinoma. *Am J Surg Pathol* 1998;22:1393–1403. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9808132>.

141. Allison KH, Love JE, Garcia RL. Epithelioid trophoblastic tumor: review of a rare neoplasm of the chorionic-type intermediate trophoblast. *Arch Pathol Lab Med* 2006;130:1875–1877. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17149967>.

142. Yang J, Zong L, Wang J, et al. Epithelioid Trophoblastic Tumors: Treatments, Outcomes, and Potential Therapeutic Targets. *J Cancer* 2019;10:11–19. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30662520>.

143. Hancock BW, Tidy J. Placental site trophoblastic tumour and epithelioid trophoblastic tumour. *Best Practice & Research Clinical*

*Obstetrics & Gynaecology* 2021;74:131–148. Available at: <https://www.sciencedirect.com/science/article/pii/S152169342030153X>.

144. Sobecki-Rausch J, Winder A, Maniar KP, et al. Surgery and Platinum/Etoposide-Based Chemotherapy for the Treatment of Epithelioid Trophoblastic Tumor. *Int J Gynecol Cancer* 2018;28:1117–1122. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/29757875>.

145. Lok C, Trommel Nv, Braicu EI, et al. Practical Guidelines for the Treatment of Gestational Trophoblastic Disease: Collaboration of the European Organisation for the Treatment of Trophoblastic Disease (EOTTD)–European Society of Gynaecologic Oncology (ESGO)–Gynecologic Cancer InterGroup (GCIg)–International Society for the Study of Trophoblastic Diseases (ISSTD). *J Clin Oncol* 2025;43:2119–2128. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/40359461>.

146. Dessources K, Aviki E, Leitao MM, Jr. Lower extremity lymphedema in patients with gynecologic malignancies. *Int J Gynecol Cancer* 2020;30:252–260. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/31915136>.

147. Bona AF, Ferreira KR, Carvalho RBM, et al. Incidence, prevalence, and factors associated with lymphedema after treatment for cervical cancer: a systematic review. *Int J Gynecol Cancer* 2020;30:1697–1704. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/32863276>.

148. Loprinzi CL, Lacchetti C, Bleeker J, et al. Prevention and Management of Chemotherapy-Induced Peripheral Neuropathy in Survivors of Adult Cancers: ASCO Guideline Update. *J Clin Oncol* 2020;38:3325–3348. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/32663120>.

149. Stahl JM, Qian JM, Tien CJ, et al. Extended duration of dilator use beyond 1 year may reduce vaginal stenosis after intravaginal high-dose-rate brachytherapy. *Support Care Cancer* 2019;27:1425–1433. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30187220>.

150. Park HS, Ratner ES, Lucarelli L, et al. Predictors of vaginal stenosis after intravaginal high-dose-rate brachytherapy for endometrial



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carcinoma. Brachytherapy 2015;14:464–470. Available at:  
<https://www.ncbi.nlm.nih.gov/pubmed/25887343>.

151. Dracham CB, Shankar A, Madan R. Radiation induced secondary malignancies: a review article. Radiat Oncol J 2018;36:85–94. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/29983028>.

152. Schepisi G, Casadei C, Toma I, et al. Immunotherapy and Its Development for Gynecological (Ovarian, Endometrial and Cervical) Tumors: From Immune Checkpoint Inhibitors to Chimeric Antigen Receptor (CAR)-T Cell Therapy. Cancers (Basel) 2021;13:840. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/33671294>.

153. Borella F, Preti M, Bertero L, et al. Is There a Place for Immune Checkpoint Inhibitors in Vulvar Neoplasms? A State of the Art Review. Int J Mol Sci 2020;22:190. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/33375467>.

154. Lin KY, Frawley HC, Denehy L, et al. Exercise interventions for patients with gynaecological cancer: a systematic review and meta-analysis. Physiotherapy 2016;102:309–319. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27553642>.

155. Nekhlyudov L, Mollica MA, Jacobsen PB, et al. Developing a Quality of Cancer Survivorship Care Framework: Implications for Clinical Care, Research, and Policy. J Natl Cancer Inst 2019;111:1120–1130. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/31095326>.

156. Bober SL, Reese JB, Barbera L, et al. How to ask and what to do: a guide for clinical inquiry and intervention regarding female sexual health after cancer. Curr Opin Support Palliat Care 2016;10:44–54. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26716390>.

157. Damast S, Jeffery DD, Son CH, et al. Literature Review of Vaginal Stenosis and Dilator Use in Radiation Oncology. Pract Radiat Oncol 2019;9:479–491. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/31302301>.

158. Campbell G, Thomas TH, Hand L, et al. Caring for Survivors of Gynecologic Cancer: Assessment and Management of Long-term and Late Effects. Semin Oncol Nurs 2019;35:192–201. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30867102>.

159. SGO/FWC Survivorship Toolkit. Available at: <https://www.sgo.org/resources/survivorship-toolkit/>. Accessed 01/31/2025.